

Scholars High School Physics 1: Mechanics (4926)

Andrew Sayler

Fri

Sep 11, 2026 - Mar 12, 2027

7:30 PM - 9:00 PM ET (4:30 PM - 6:00 PM PT)

Overview

Week 1 (Sep 11) Class Transcript - Velocity



[Go back to the class overview page](#)

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aninconle 2026-09-11 19:24:57

Scholars High School Physics 1: Mechanics

aninconle 2026-09-11 19:25:02

Hi 

Zip-a-Dee-Doo-Dah 2026-09-11 19:25:19

Hello!

veryhungrypanda 2026-09-11 19:25:19

hellooo 

aninconle 2026-09-11 19:25:24

The classroom is now **moderated**. This means that the messages you type will come to the instructors rather than going directly into the room. We'll choose some of the messages to share with all the students.

hw_zasf.xAx 2026-09-11 19:25:29

hello!

Canada1334 2026-09-11 19:25:29

Hello 

aninconle 2026-09-11 19:25:35

Please feel free to **ask questions at any time** to get help from our teaching staff! Let's introduce them now.

aninconle 2026-09-11 19:25:37

My name is **Andrew Sayler** and I am the instructor for this course.

aninconle 2026-09-11 19:25:38

With Bachelor's degrees in Mathematics and Physics, Andrew Sayler joined AoPS in October 2016. His love for math was ignited by its puzzle-solving nature. He has over a decade of experience teaching students from elementary school to the college level, including performing physics demonstrations to inspire children's interest in the field. Andrew loves creating a discovery-based environment where students can discover math through rigorous learning and guided exploration. His pedagogical approach is centered around the excitement of seeing students light up as they grasp new concepts. Outside the classroom, Andrew is engrossed in reading, exploring puzzles, and diving into the histories of words and recipes. He enjoys board games, building outdoor ovens for cooking, and spending quality time with his family, embracing every moment of shared learning and fun.

aninconle 2026-09-11 19:25:43

Your assistants for this course will be **Angel Adrian Rojas Jimenez** (Angel.Rojas) and **Lee Trent** (vacuously-true).

aninconle 2026-09-11 19:25:45

Angel.Rojas: Angel Rojas is a math and coding enthusiast with a Bachelor's degree in Math. During his university studies, he created the Math Club at his alma mater, Yachay Tech in Ecuador. Also, he participated in several Ecuadorian and Peruvian math

Olympiads at the middle school, high school, and university levels. Now, he is working at Wolfram|Alpha LLC as a Technical Analyst and is a teacher of Programming at an Ecuadorian High School. The most valuable part of his professional career in education is to inspire students by answering their questions and making them intrigued about the unknown.

aninconle 2026-09-11 19:25:49

vacuously-true: Lee Trent is a devoted math educator with a rich and varied background, including classroom teaching from elementary school to the university level, working at high school math camps for many summers, and writing problems for the Minnesota State High School Math Competition. She holds a B.S. in Mathematics with minors in Astronomy, Political Science, and German, and is currently pursuing a Ph.D. in Mathematics at the University of Minnesota. She also helped to found a conference to help undergraduate math students learn about potential career paths in mathematics. Lee loves seeing her students make their own mathematical discoveries, and she strives to give her students an active role in their own learning by creating a collaborative environment and meeting them where they are. When she's not teaching, Lee enjoys painting, riding her bike, playing board games, and hanging out with her cats.

Angel.Rojas 2026-09-11 19:25:53

Hi everyone! 🤖

Canada1334 2026-09-11 19:26:16

hey! 😊

Zip-a-Dee-Doo-Dah 2026-09-11 19:26:16

Hello Angel Rojas!

kabir_sahil 2026-09-11 19:26:25

hi

guacataj 2026-09-11 19:26:25

hii

Ash_Ketchup13 2026-09-11 19:26:30

hi

Nebula233 2026-09-11 19:26:30

hai

flybirdeee_ 2026-09-11 19:26:30

hi

aninconle 2026-09-11 19:26:33

If you want the classroom to stop auto-scrolling so that you can read all those bios, scroll up a little bit. Make sure to scroll back down when you're done so that the classroom will resume auto-scrolling.

aninconle 2026-09-11 19:26:37

We post "sticky" problems and important comments to the top of the room.

aninconle 2026-09-11 19:26:38

You can slide the bar between the "sticky" portion of the room on the top of the classroom and the rest of the classroom to make it larger or smaller. Alternatively, click the arrow to open or close the "sticky."

aninconle 2026-09-11 19:26:40

You can change the classroom size, font size, colors, and other settings in the Settings Menu at the top-right of your screen. Another fun option is the style of your emojis. Send us an emoji to celebrate our first lesson!

aninconle 2026-09-11 19:26:45

One of the many benefits of our classroom is that dozens of you can ask questions or share ideas simultaneously, so please participate.

aninconle 2026-09-11 19:26:48

You'll know that we received your message if you see "sent to instructor." We value every message you send us, even if we don't directly respond.

aninconle 2026-09-11 19:26:51

Often we will send you individual responses to questions or comments as a **whisper**.

aninconle 2026-09-11 19:26:55

Sometimes, we'll solicit your responses using fancy technology like this.

Select a button and then click SUBMIT ANSWER.

Send me a demonstration whisper, please!

I just like clicking buttons.

I don't want a whisper, thank you.

aninconle 2026-09-11 19:27:11

We also might pass your questions into the classroom so the entire class can benefit from the discussion.

aninconle 2026-09-11 19:27:14

Students can use $LATEX$ throughout the site, although $LATEX$ is *never required*.

aninconle 2026-09-11 19:27:17

If you happen to already know $LATEX$, send us a quick example!

aninconle 2026-09-11 19:27:21

If you don't know $LATEX$ and want to start learning, for most mathematical notation you can simply begin and end it with a dollar sign.

aninconle 2026-09-11 19:27:26

For example,

$$2^2=4.$$

displays $2^2 = 4$.

Zip-a-Dee-Doo-Dah 2026-09-11 19:27:31

$$\sin(C) = \frac{x(\sqrt{3} + \sqrt{35})/2}{6x}$$

VeryDragon2013 2026-09-11 19:27:43

$LATEX$ is great

anisquash 2026-09-11 19:27:47

$$\sqrt{610100}$$

veryhungrypanda 2026-09-11 19:27:49

$$6^7 = 67.$$

aninconle 2026-09-11 19:27:51

You can check whether your $LATEX$ is behaving as expected by clicking the eye, , to preview your message before sending it.

aninconle 2026-09-11 19:27:54

If you've already sent a message and want to tweak it, click [CTRL]-[UP] (or [CMD]-[UP]) to re-populate the typing box with a prior message.

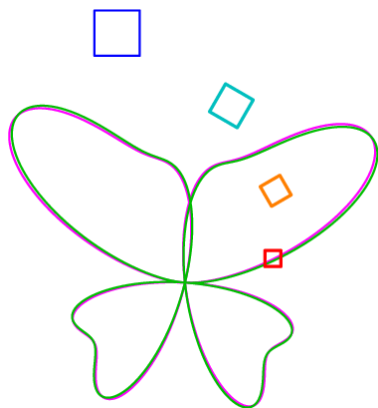
aninconle 2026-09-11 19:27:58

If you want to learn more complicated $LATEX$, check out the [Resources](#) section of our website, or ask us for help during class or on the message board.

aninconle 2026-09-11 19:27:59

We'll also show images in the classroom:

aninconle 2026-09-11 19:28:01



aninconle 2026-09-11 19:28:04

We'll provide a full transcript after the lesson so you can review what happened, so you do *not* need to take notes in real time.

aninconle 2026-09-11 19:28:07

You'll still want pencil and paper for your scratch work, as we will solve the problems together.

Pencil ready!

Pen ready!

Chalk ready!

Marker ready!

Be right back!

aninconle 2026-09-11 19:28:41

Also, for my curiosity, have you taken an AoPS class before?

Nope

Yep, one

Many

aninconle 2026-09-11 19:29:59

If you're just coming in now and are new to AoPS Online, scroll up to see the instructions on how to use the classroom. You can also ask an assistant for help.

aninconle 2026-09-11 19:30:40

Alright! Let's get into some physics.

aninconle 2026-09-11 19:30:42

Scholars High School Physics 1: Mechanics Lesson 1: Velocity

aninconle 2026-09-11 19:30:46

Mechanics

aninconle 2026-09-11 19:31:01

Can you think of some things we might study in mechanics? Go ahead and type one or two.

Ash_Ketchup13 2026-09-11 19:31:55

motion

Ash_Ketchup13 2026-09-11 19:31:55

torque

M.Zhang555 2026-09-11 19:31:55

motion

Potatomiaomi 2026-09-11 19:31:55

motion?

M.Zhang555 2026-09-11 19:31:55

forces

mocha_x 2026-09-11 19:31:55

gears torque power stuff?

aahelic13 2026-09-11 19:31:55

motion

Nebula233 2026-09-11 19:31:55

I'm guessing velocity cuz it says lesson 1 velocity?

emilywow 2026-09-11 19:31:55

maybe like motion

kanishka.chittoor 2026-09-11 19:31:55

acceleration

flybirdeee_ 2026-09-11 19:31:55

forces

anisquash 2026-09-11 19:31:55

Energy and forces

M.Zhang555 2026-09-11 19:31:55

kinematics

LCLARK5691 2026-09-11 19:31:55

fluids

flybirdeee_ 2026-09-11 19:31:55

gravity

aahelic13 2026-09-11 19:31:55

rotational stuff

Thorongil 2026-09-11 19:31:55

moving objects (balls, blocks, arrows, rockets,...)

AN29 2026-09-11 19:31:55

forces, momentum

pandafrog 2026-09-11 19:31:55

friction

pandafrog 2026-09-11 19:31:55

gravity

Ash_Ketchup13 2026-09-11 19:31:55

centrifugal and centripetal forces

M.Zhang555 2026-09-11 19:31:55

energy

VeryDragon2013 2026-09-11 19:31:55

kinetic and potential energy

guacataj 2026-09-11 19:31:55

hmmm, maybe velocity and acceleration

aninconle 2026-09-11 19:32:01

These are great! In mechanics, we study all types of motion.

aninconle 2026-09-11 19:32:03

Scientists and engineers always have questions.

aninconle 2026-09-11 19:32:05

For example:

- Does adding more blades to a windmill let it generate more electricity?

- How long does an airport runway need to be for planes to take off?

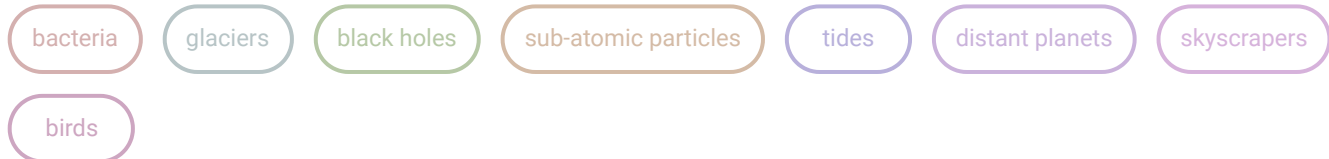
- How long will it take a rocket to go from Earth to Mars?

aninconle 2026-09-11 19:32:11

These are all practical questions, and we need to understand motion to answer them.

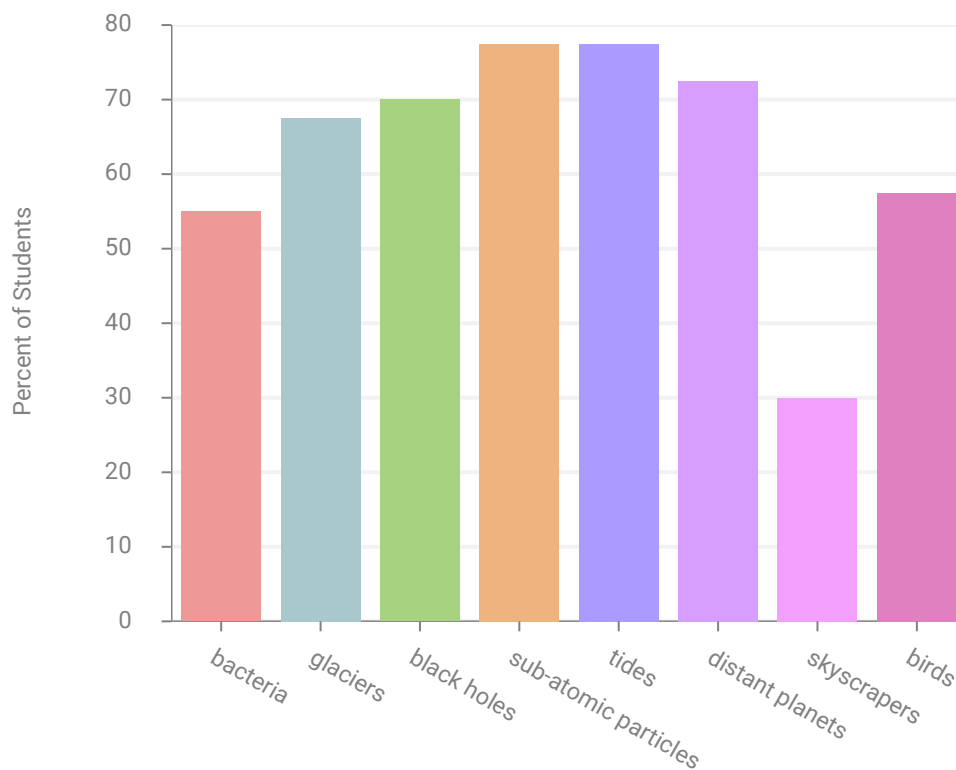
aninconle 2026-09-11 19:32:15

Which of these are things you'd expect scientists to study the motion of?



aninconle 2026-09-11 19:33:32

Which of these are things you'd expect scientists to study the motion of?



aninconle 2026-09-11 19:33:36

Scientists study the motion of all those things. Mechanics can be applied to many different fields of study!

aninconle 2026-09-11 19:33:37

Our first task is to describe motion *quantitatively*. This means using numbers, graphs, and equations to explain motion.

aninconle 2026-09-11 19:33:39

Today we're going to study speed, velocity, coordinates, graphs. These are all ideas. I'll sometimes call them "tools." They're not physical tools like a hammer; they're conceptual tools. When we combine them to understand a phenomenon, we've built a *conceptual model*.

aninconle 2026-09-11 19:33:41

A Rental Car: Speed, Units, and Sense-Making

aninconle 2026-09-11 19:33:50

In August 2022, a woman in Vancouver rented a car for three days. The rental company billed her more than \$9,000. They said the rate is $\frac{\$0.25}{\text{km}}$, and that the woman drove 36,482 km. If you were the renter, what would you have said in reply?

pandafrog 2026-09-11 19:34:40

That the price is correct,

M.Zhang555 2026-09-11 19:34:40

thats impossible to drive 36482 km in 3 days bro

anisquash 2026-09-11 19:34:40

This is super expensive!

askuo 2026-09-11 19:34:40

i wouldve said thats totally fair but thats expensive as hell

clairexc 2026-09-11 19:34:40

"Can I get a discount?"

flybirdeee_ 2026-09-11 19:34:40

u wasted the money!

Nebula233 2026-09-11 19:34:40

that's like way too expensive

veryhungrypanda 2026-09-11 19:34:45

THAT DOESNT SOUND RIHGT I WANNA SPEAK TO YOUR MANAGER

aahelic13 2026-09-11 19:34:45

That makes sense

Ash_Ketchup13 2026-09-11 19:34:45

That's way too much to charge somebody.

Zip-a-Dee-Doo-Dah 2026-09-11 19:34:45

You're wrong,

aninconle 2026-09-11 19:34:47

The arithmetic is okay. $(36,482 \text{ km}) \cdot \left(\frac{\$0.25}{\text{km}}\right) = \$9,120.5$. **That's not exactly what the problem says, but it's approximately correct.**

aninconle 2026-09-11 19:34:49

What's wrong here?

pandafrog 2026-09-11 19:36:13

but the distance is wrong.

clairexc 2026-09-11 19:36:13

You can't drive that much

Ash_Ketchup13 2026-09-11 19:36:13

its basically impossible to drive 36482 km in 3 days

mocha_x 2026-09-11 19:36:13

There's no way she drove 36,482km in 3 days

AwesomeAlligator57 2026-09-11 19:36:13

36482 km is a lot for only 3 days....

syl314 2026-09-11 19:36:13

Driving 36000 kilometers in 3 days

Diamond_Zombie67 2026-09-11 19:36:13

i did NOT drive 36k kilometers

aninconle 2026-09-11 19:36:16

36,482 km is really far! Can you think of a calculation that would show the rental car company we couldn't have driven that far?

VeryDragon2013 2026-09-11 19:37:54

speed

Exponent11 2026-09-11 19:37:54

speed?

M.Zhang555 2026-09-11 19:37:54

use average speed

AllenLuo 2026-09-11 19:37:54

convert it to like km/hr

flybirdeee_ 2026-09-11 19:37:54

what the max speed the car can drive is and use that to like see the maximum distance the woman could have driven in a day

Nebula233 2026-09-11 19:37:54

find speed you would need to drive at through those 3 days

M.Zhang555 2026-09-11 19:37:54

because then the average speed is not possible

mocha_x 2026-09-11 19:37:54

we can see the km/hr

aninconle 2026-09-11 19:37:56

Let's find the car's average speed over those three days.

aninconle 2026-09-11 19:37:58

To find speed, we divide what by what?

Bunnicorn 2026-09-11 19:38:40

distance/time

emilywow 2026-09-11 19:38:40

divide distance by time

Nebula233 2026-09-11 19:38:40

distance divided by time

anisquash 2026-09-11 19:38:40

distance divided by timee

Oceanix 2026-09-11 19:38:40

distacne/time

kabir_sahil 2026-09-11 19:38:40

distance/time taken

TatuShark 2026-09-11 19:38:40

distance by time

Thorongil 2026-09-11 19:38:40

distance by time

apollo2000 2026-09-11 19:38:40

distance/time

WildOctopus 2026-09-11 19:38:40

distance over time

Potatomiaomi 2026-09-11 19:38:40

distance by time

AllenLuo 2026-09-11 19:38:40

Distance/Time

AN29 2026-09-11 19:38:40

distance/time

aahelic13 2026-09-11 19:38:40

distance/time

Ash_Ketchup13 2026-09-11 19:38:40

distance/time

askuo 2026-09-11 19:38:40

distance over time

Seifr 2026-09-11 19:38:49

distance/time

Brightgiraffe 2026-09-11 19:38:49

distance divided by time

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We divide a distance traveled by a time.

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The division here gives

$$v = \frac{36,482 \text{ km}}{3 \text{ d}}$$

$$= 12,160.667 \frac{\text{km}}{\text{d}}.$$

aninconle 2026-09-11 19:38:56

The symbol km is for kilometers and d is for days. These and all other units in this course will be in upright font, for example km instead of *km*.

aninconle 2026-09-11 19:38:57

***v* is our symbol for speed.**

aninconle 2026-09-11 19:39:04

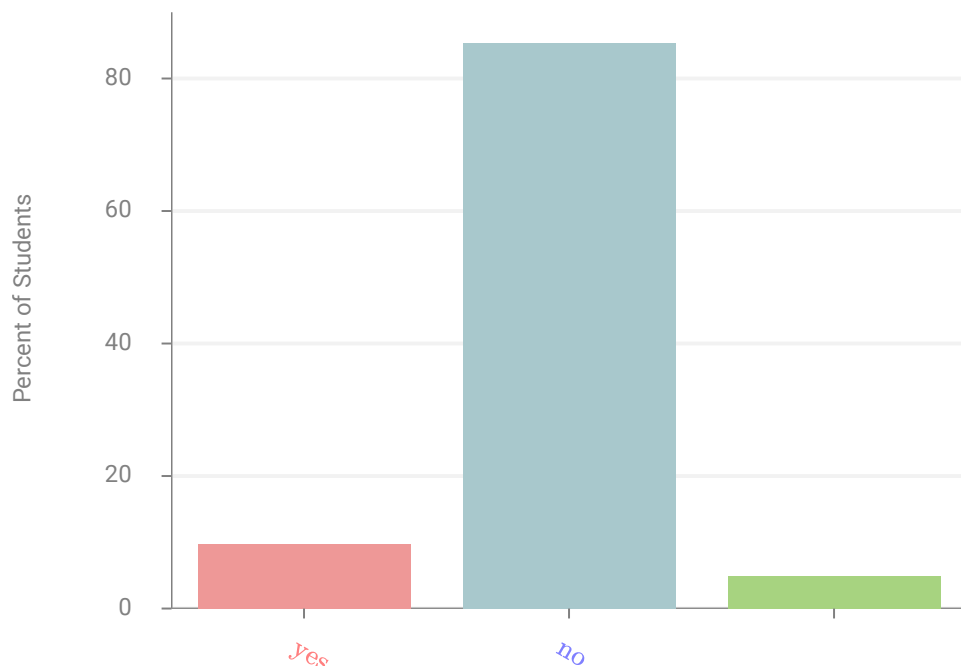
Do we need that extra .667 in 12,160.667, given what we're calculating?

yes

no

I'm not sure how to decide

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aninconle 2026-09-11 19:40:06

We don't care about the exact speed. We just want to know if it's reasonable. Let's round to $v \approx 12,000 \frac{\text{km}}{\text{d}}$.

aninconle 2026-09-11 19:40:19

We're using $\approx$, not $=$, to remind us that we're rounding and don't need precise values.

aninconle 2026-09-11 19:40:22

Technically, $\frac{36,482 \text{ km}}{3\text{d}} \approx 12,160.667 \frac{\text{km}}{\text{d}}$, since really it's $12,160.6666 \dots$, with a repeating decimal, if we want the two quantities to be exactly equal.

aninconle 2026-09-11 19:40:30

We didn't use an $\approx$ sign in $\frac{36,482 \text{ km}}{3\text{d}} = 12,160.667 \frac{\text{km}}{\text{d}}$ because although it's technically an approximation, we weren't thinking about it as an approximation yet. That will be a pattern in this class. We'll often write out the decimal for something and use an equal sign, even if it's not exactly equal. And we'll try to use approximation signs when we're intentionally treating something as an estimate.

aninconle 2026-09-11 19:40:33

Our estimate $v \approx 12,000 \frac{\text{km}}{\text{d}}$ is a speed, but it's not written in a way that's very close to everyday experience. What can we do to make it more intuitive to the car rental company just how fast this speed is?

AN29 2026-09-11 19:41:28

change the days to hours

VeryDragon2013 2026-09-11 19:41:28

convert days to hours so km/hr

emilywow 2026-09-11 19:41:28

divide by 24 to make it km/hr

pandafrog 2026-09-11 19:41:28

km per hour

syl314 2026-09-11 19:41:28

Change it to km/hr

M.Zhang555 2026-09-11 19:41:28

make it per hour

Bunnicorn 2026-09-11 19:41:28

convert it to km/hr

AllenLuo 2026-09-11 19:41:28

km/hour

aahelic13 2026-09-11 19:41:28

per hour instead of per day

rysaishi 2026-09-11 19:41:28

change it to km/h

guacataj 2026-09-11 19:41:28

write it in km/hr

nickypicky39084 2026-09-11 19:41:28

kph

Canada1334 2026-09-11 19:41:28

km per hour instead of days

aninconle 2026-09-11 19:41:29

Let's convert to more common units.

aninconle 2026-09-11 19:41:30

What's the speed in km/h ?

$$12,000 \frac{\text{km}}{\text{d}} = \boxed{} \frac{\text{km}}{\text{h}}.$$

M.Zhang555 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

kanishka.chittoor 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

clairexc 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

Diamond_Zombie67 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

rysaishi 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

Exponent11 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

emilywow 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

CodeWrecker 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

anisquash 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

mocha_x 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

AN29 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

CAAI 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

guacataj 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

Zip-a-Dee-Doo-Dah 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

LCLARK5691 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

loripachiardi 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

Bunnicorn 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

VeryDragon2013 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

agoel2010 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

WildOctopus 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

veryhungrypanda 2026-09-11 19:42:53

$$12,000 \frac{\text{km}}{\text{d}} = 500 \frac{\text{km}}{\text{h}}.$$

aninconle 2026-09-11 19:42:58

Here's a systematic way.

aninconle 2026-09-11 19:43:00

We start with

$$1 \text{ d} = 24 \text{ h}.$$

aninconle 2026-09-11 19:43:01

Dividing both sides by 24 h, we get

$$\frac{1 \text{ d}}{24 \text{ h}} = 1.$$

aninconle 2026-09-11 19:43:03

This is a *conversion factor*.

aninconle 2026-09-11 19:43:05

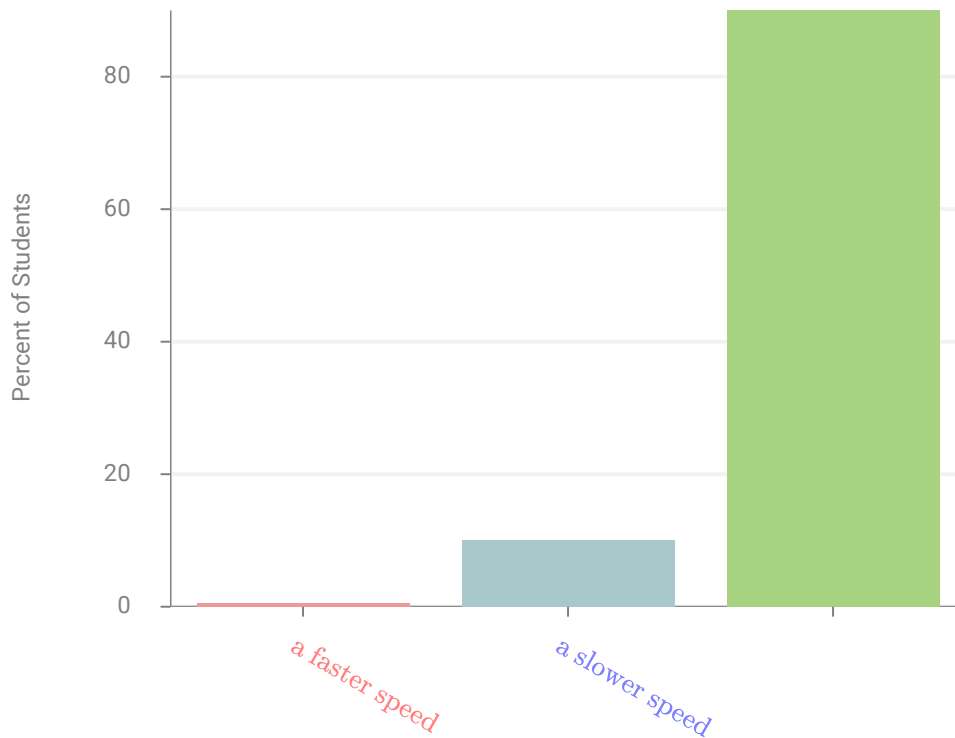
Since $\frac{1 \text{ d}}{24 \text{ h}} = 1$, what do we get when we multiply v by $\frac{1 \text{ d}}{24 \text{ h}}$?

a faster speed

a slower speed

the same speed, but with different units

aninconle 2026-09-11 19:43:52



aninconle 2026-09-11 19:43:59

Whatever v is, $v \cdot 1 = v$. The conversion factor only changes the units.

aninconle 2026-09-11 19:44:07

The unit conversion is

$$v = (v) \cdot (1) \approx \left(12,000 \frac{\text{km}}{\text{d}}\right) \cdot \left(\frac{1 \text{ d}}{24 \text{ h}}\right).$$

aninconle 2026-09-11 19:44:10

The days (d) cancel out. We're left with

$$v \approx \left(12,000 \frac{\text{km}}{\text{d}}\right) \cdot \left(\frac{1 \text{ d}}{24 \text{ h}}\right)$$

$$= 500 \frac{\text{km}}{\text{h}}.$$

aninconle 2026-09-11 19:44:14

There's an $\approx$ after v because we're working with an approximation for the speed, but there's an $=$ between

$\left(12,000 \frac{\text{km}}{\text{d}}\right) \cdot \left(\frac{1 \text{ d}}{24 \text{ h}}\right)$ and $500 \frac{\text{km}}{\text{h}}$ because those quantities are exactly equal.

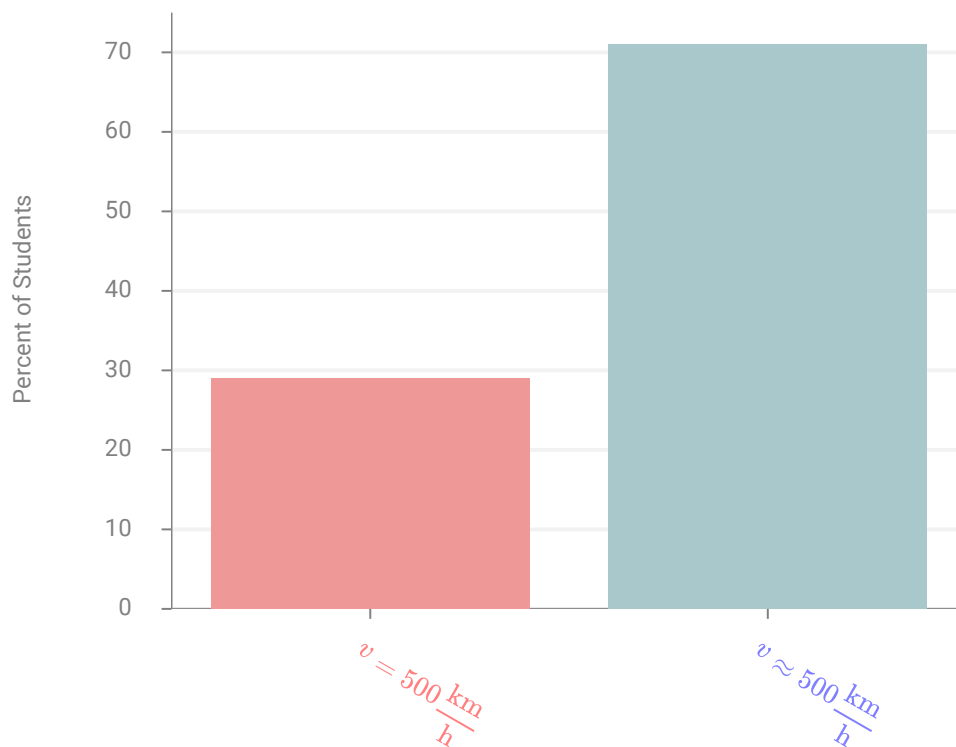
aninconle 2026-09-11 19:44:16

If we want to relate v and $500 \frac{\text{km}}{\text{h}}$ directly, which of these should we use?

$v = 500 \frac{\text{km}}{\text{h}}$

$v \approx 500 \frac{\text{km}}{\text{h}}$

aninconle 2026-09-11 19:45:10



aninconle 2026-09-11 19:45:16

We should use $v \approx 500 \frac{\text{km}}{\text{h}}$ because we're still approximating v . The unit conversion was exact, but the quantity we were converting in the first place was approximate, so the result is still approximate.

aninconle 2026-09-11 19:45:19

Next, let's convert to the US unit miles per hour, $\frac{\text{mi}}{\text{h}}$ (also written "mph")

aninconle 2026-09-11 19:45:21

There are several approximate conversions between kilometers and miles. Of these, which will make the arithmetic easiest when we turn it into a conversion factor?

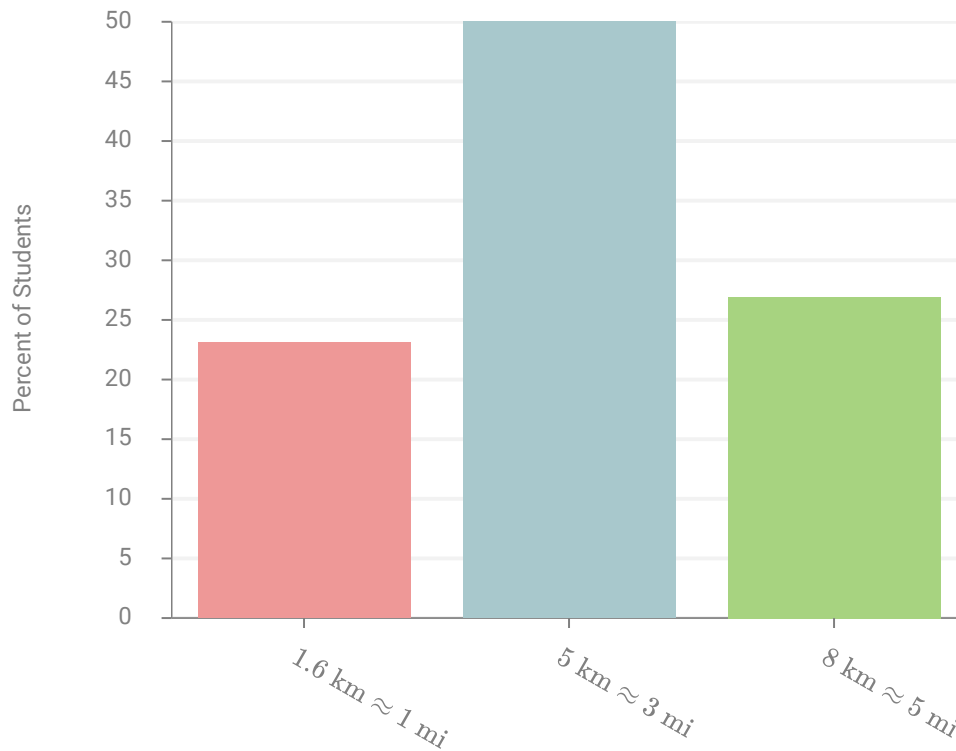
$$1.6 \text{ km} \approx 1 \text{ mi}$$

$$5 \text{ km} \approx 3 \text{ mi}$$

$$8 \text{ km} \approx 5 \text{ mi}$$

aninconle 2026-09-11 19:46:09

There are several approximate conversions between kilometers and miles. Of these, which will make the arithmetic easiest when we turn it into a conversion factor?



aninconle 2026-09-11 19:46:13

Our starting point is $v \approx 500 \frac{\text{km}}{\text{h}}$, and we want to get rid of the kilometers. Our conversion factor should have km in the denominator.

aninconle 2026-09-11 19:46:15

If we put 5 km in the denominator, it will divide the 500 km in our numerator nicely. Let's use

$$1 \approx \frac{3 \text{ mi}}{5 \text{ km}}$$

aninconle 2026-09-11 19:46:17

What is v in $\frac{\text{mi}}{\text{h}}$?

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx \boxed{} \frac{\text{mi}}{\text{h}}$$

syl314 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Oceanix 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Diamond_Zombie67 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

apollo2000 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

CodeWrecker 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

kabir_sahil 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

LCLARK5691 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

AllenLuo 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Seifr 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

thekitsune 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

agoel2010 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

clairexc 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Marshall_Huang 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Canada1334 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Ash_Ketchup13 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

pandafrog 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

TatuShark 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Potatomiaomi 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

aahelic13 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

hw_zasf.xAx 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

Nebula233 2026-09-11 19:48:04

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

nickypicky39084 2026-09-11 19:48:07

$$v \approx 500 \frac{\text{km}}{\text{h}} \approx 300 \frac{\text{mi}}{\text{h}}$$

aninconle 2026-09-11 19:48:11

$$\begin{aligned} v &\approx \left(500 \frac{\text{km}}{\text{h}} \right) \cdot \left(\frac{3 \text{ mi}}{5 \text{ km}} \right) \\ &\approx 300 \frac{\text{mi}}{\text{h}}. \end{aligned}$$

aninconle 2026-09-11 19:48:15

What would you say to the rental company?

Canada1334 2026-09-11 19:49:55

I dont drive an F1 car

hw_zasf.xAx 2026-09-11 19:49:55

I was not going 300 mph

VeryDragon2013 2026-09-11 19:49:55

I NEED TO TALK TO THE MANAGER!!! I CANT HAVE GONE THAT FAR

AN29 2026-09-11 19:49:55

it is not possible for my average speed to be 300 mph

kanishka.chittoor 2026-09-11 19:49:55

impossible to go thnat fast

emilywow 2026-09-11 19:49:55

i couldnt have driven 300mph for 3 days straight

M.Zhang555 2026-09-11 19:49:55

haha thats the faster than the speed of an f1 car

Ash_Ketchup13 2026-09-11 19:49:55

Stop trying to scam me

Seifr 2026-09-11 19:49:55

An speed of 300 miles per hour is absurd even as a maximum speed, and near impossible as an average speed.

WildOctopus 2026-09-11 19:49:55

that they are trying to cheat your money

Zip-a-Dee-Doo-Dah 2026-09-11 19:49:55

Also why did you let a minor rent a car .

cicitheduckyyy 2026-09-11 19:49:55

"who drives at 300mph?"

guacataj 2026-09-11 19:49:55

The max speed of this car is less than 300 MPH, so it is statistically impossible for her to be driving for 72 hours straight at 300 MPH, hence, you overcharged her

Thorongil 2026-09-11 19:49:55

This is kind of absurd

Bunnicorn 2026-09-11 19:49:55

I may or may not have ignored the speed limit signs...

BananaPenguin 2026-09-11 19:49:55

I did not drive 36000 km

flybirdeee_ 2026-09-11 19:50:07

unless im tweaking a car cnt drive 300 mph

anisquash 2026-09-11 19:50:07

It is impossible for anyone to drive at that speed for 72 hours straight, even an airplane can't last at that much constant speed for so long.

aninconle 2026-09-11 19:50:10

The company's numbers mean we drove an average speed of 300 mi/h for three days. That's about twice as fast as race cars, and doesn't even factor in any breaks.

aninconle 2026-09-11 19:50:18

The company probably charged for the total distance the car had ever been driven. 36,482 km is a bit less than Earth's circumference, so the car had gone almost far enough to wrap around the equator.

aninconle 2026-09-11 19:50:21

In this example, what was the value of doing the unit conversions?

Thorongil 2026-09-11 19:51:48

we were able to get a realistic sense of how fast she would have to be going in order to drive that far

AllenLuo 2026-09-11 19:51:48

We could make some things more obvious

CodeWrecker 2026-09-11 19:51:48

we are able to turn the numbers into something more useful

AN29 2026-09-11 19:51:48

changing it to more understandable daily life units

Canada1334 2026-09-11 19:51:48

It changed the speed to a more familiar metric

aninconle 2026-09-11 19:51:49

Unit conversions made the numbers easier to understand. When we take a mathematical result and find ways to understand its meaning better, we're sense-making. There are many different ways to do sense-making, so look out for more examples!

aninconle 2026-09-11 19:51:52

McGowan Rides Uphill: Estimating a speed

aninconle 2026-09-11 19:52:05

Use a video clip to estimate a cyclist's speed.

aninconle 2026-09-11 19:52:07

Here's professional cyclist Ayesha McGowan riding up a hill:

aninconle 2026-09-11 19:52:08



aninconle 2026-09-11 19:52:18

Another approximate conversion is $1 \text{ m/s} \approx 2 \text{ mi/h}$.

aninconle 2026-09-11 19:52:22

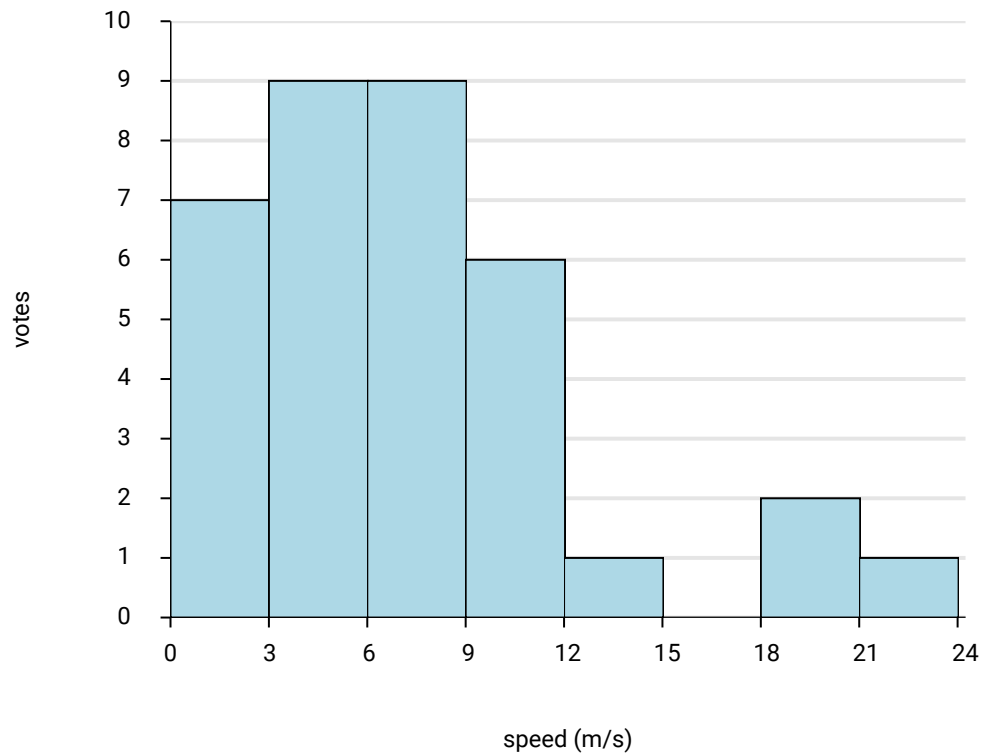
What's your initial guess on how fast McGowan is going?

McGowan's speed $\approx$ m/s

aninconle 2026-09-11 19:53:50

Here's a *histrogram* of our results:

aninconle 2026-09-11 19:53:57



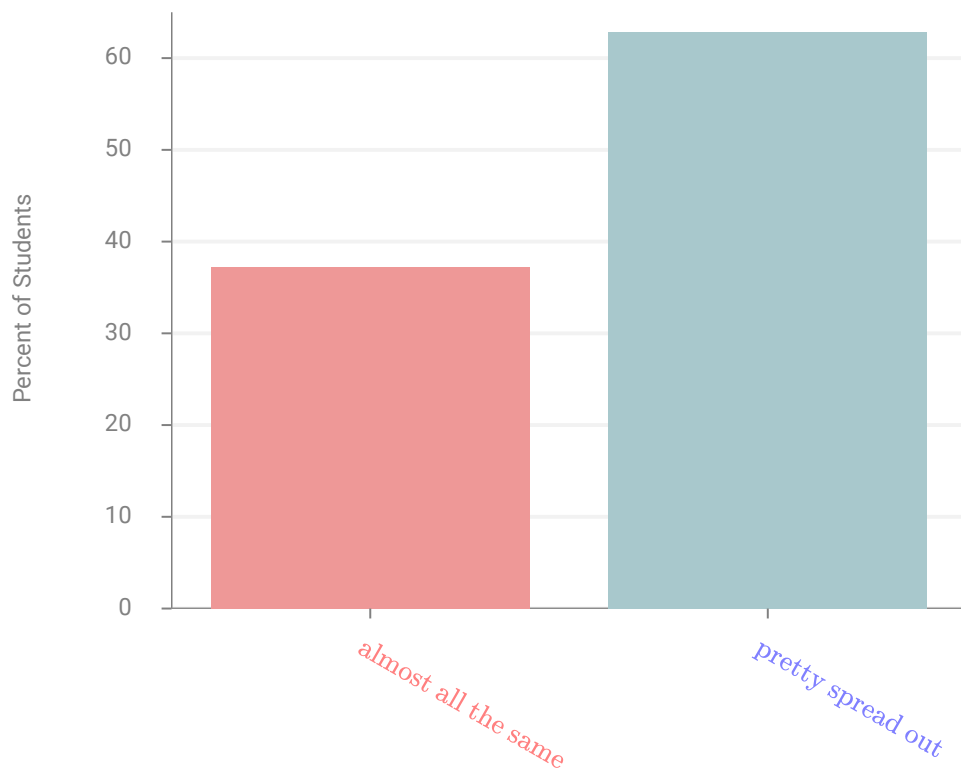
aninconle 2026-09-11 19:54:09

Would you say our estimates are...

almost all the same

pretty spread out

aninconle 2026-09-11 19:55:16



aninconle 2026-09-11 19:55:18

It's hard to just guess the speed. In physics, when we estimate, we don't just take a guess. We break the problem down, using whatever information we can find.

aninconle 2026-09-11 19:55:20

Do you see anything in the video we could use to improve our estimate, assuming we can make measurements on the video?

cicitheduckyyy 2026-09-11 19:57:00

maybe like size of her bike's wheels?

AllenLuo 2026-09-11 19:57:00

lots of trees in the background

thekitsune 2026-09-11 19:57:00

time of video

mocha_x 2026-09-11 19:57:00

we can use the background as a reference

Oceanix 2026-09-11 19:57:00

The tire diameter approximately

vmene 2026-09-11 19:57:00

the background

vmene 2026-09-11 19:57:00

the camera speed

AN29 2026-09-11 19:57:00

we could find distance and time to find the speed

kabir_sahil 2026-09-11 19:57:00

how long it was?

syl314 2026-09-11 19:57:00

The average height of a human?

WildOctopus 2026-09-11 19:57:00

how fast the wheel spins

flybirdeee_ 2026-09-11 19:57:00

the distance the guy traveled

Marshall_Huang 2026-09-11 19:57:00

video is made of a bunch of photos, compare time between photos with distance part of the bike moved

guacataj 2026-09-11 19:57:00

a stretch of road

Sefr 2026-09-11 19:57:00

The length of the bike, signpost, or nearby foliage.

Nebula233 2026-09-11 19:57:00

time it and mark the starting place and the place where the cyclist is after 1 second. or js time whole thing and measure distance and divide by # of seconds the video lasted

syl314 2026-09-11 19:57:00

And the average height of a bike?

guacataj 2026-09-11 19:57:00

and the bike size

Canada1334 2026-09-11 19:57:00

Measure the time and maybe distance traveled if you use the bike as a reference you can find around how many meters

Zip-a-Dee-Doo-Dah 2026-09-11 19:57:00

The ground, and cracks in the pavement?

kanishka.chittoor 2026-09-11 19:57:00

wheel rotations

AwesomeAlligator57 2026-09-11 19:57:00

Seeing how tall McGowan is and using her height as a reference to length

apollo2000 2026-09-11 19:57:00

she passes over the region between the cracks in the sidewalk in about 1 second, so if we measure that we can estimate her speed.

Diamond_Zombie67 2026-09-11 19:57:00

ayesha's body, the circumference of the wheels for scale, maybe the background too but probably not

aahelic13 2026-09-11 19:57:00

seeing how long the video was and around how many meters we think she rode

aninconle 2026-09-11 19:57:02

To measure speed, we need to find both a distance and a time.

aninconle 2026-09-11 19:57:04

To get time, we can use the frames of the video.

aninconle 2026-09-11 19:57:07

The video is made of individual frames pasted together. Here's the 18th frame:

aninconle 2026-09-11 19:57:08



aninconle 2026-09-11 19:57:16

The video plays at 20 frames per second, so $\frac{1}{20}$ s passes between frames.

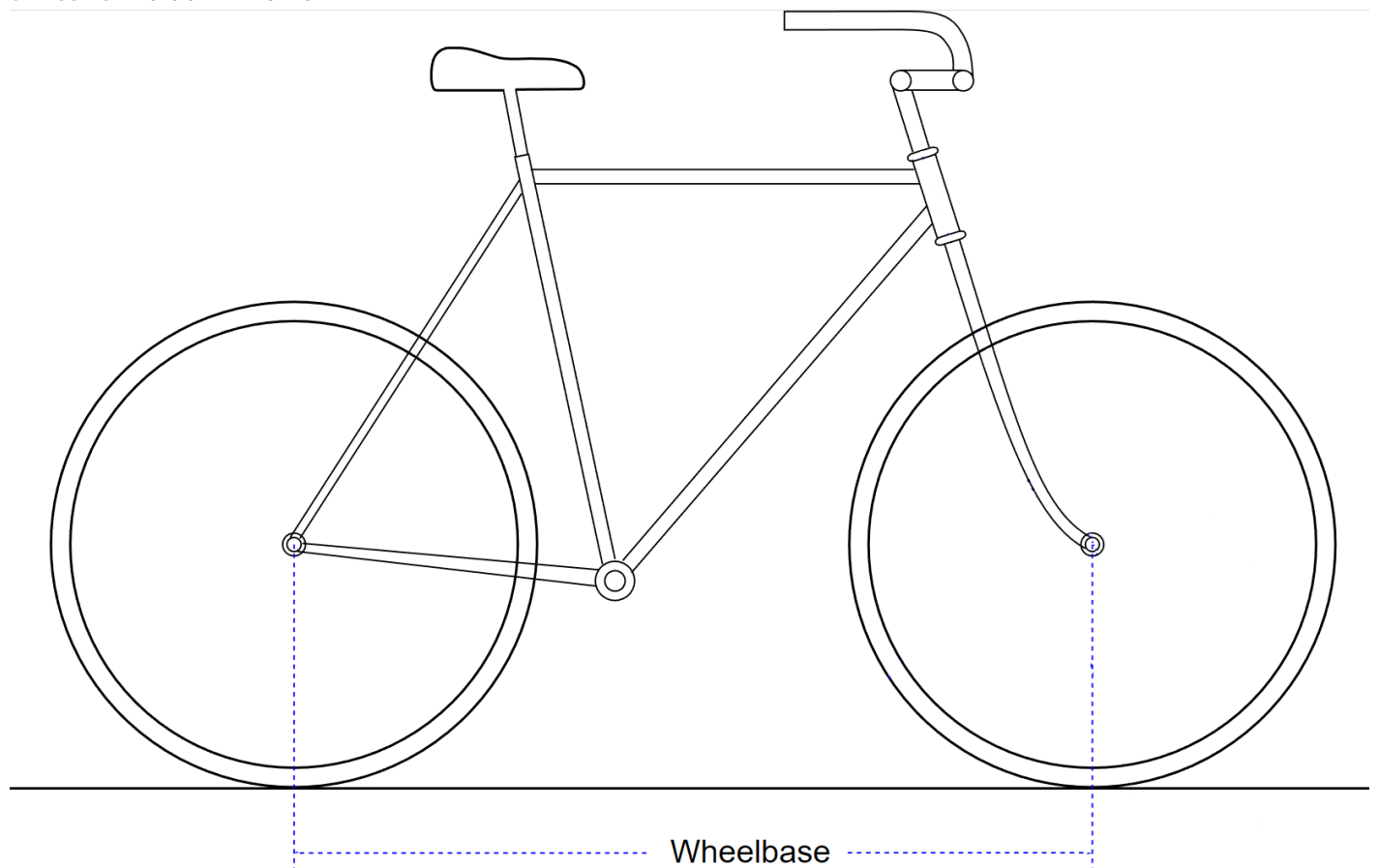
aninconle 2026-09-11 19:57:21

For distances, we want an object whose size we know. We can use that object as a reference. This is called finding a *scale*.

aninconle 2026-09-11 19:57:24

The *wheelbase* of the bike is a good scale.

aninconle 2026-09-11 19:57:26



aninconle 2026-09-11 19:57:29

McGowan's sponsor says her wheelbase is 0.977 m.

aninconle 2026-09-11 19:57:36

Here's frame 18 again.

aninconle 2026-09-11 19:57:37



aninconle 2026-09-11 19:57:40

What might happen in a few frames that we can use to measure how far McGowan has gone?

Nova_Wu_2013 2026-09-11 19:59:18

how far the wheelbase has traveled

apollo2000 2026-09-11 19:59:18

She might pass over another crack in the sidewalk.

Zip-a-Dee-Doo-Dah 2026-09-11 19:59:18

The back of the wheelbase going over the crack in the pavement.

pandafrog 2026-09-11 19:59:18

how quickly the crack in the road goes

flybirdeee_ 2026-09-11 19:59:18

how many wheelbases she traveled

CodeWrecker 2026-09-11 19:59:18

her distance relative to the cracks in the road will change

Thorongil 2026-09-11 19:59:18

we see how many frames it takes for McGowan's back axle to reach that crack in the road where her front axle is right now

aninconle 2026-09-11 19:59:21

McGowan's front wheel is touching a crack in the road. Let's look ahead a couple frames until her back wheel is on the crack.

aninconle 2026-09-11 19:59:23

Here's frame 20:

aninconle 2026-09-11 19:59:24



aninconle 2026-09-11 19:59:28

Below, I've highlighted the crack in red (line 2) and edges of the wheelbase in green (lines 1 and 3).

aninconle 2026-09-11 19:59:29



aninconle 2026-09-11 19:59:34

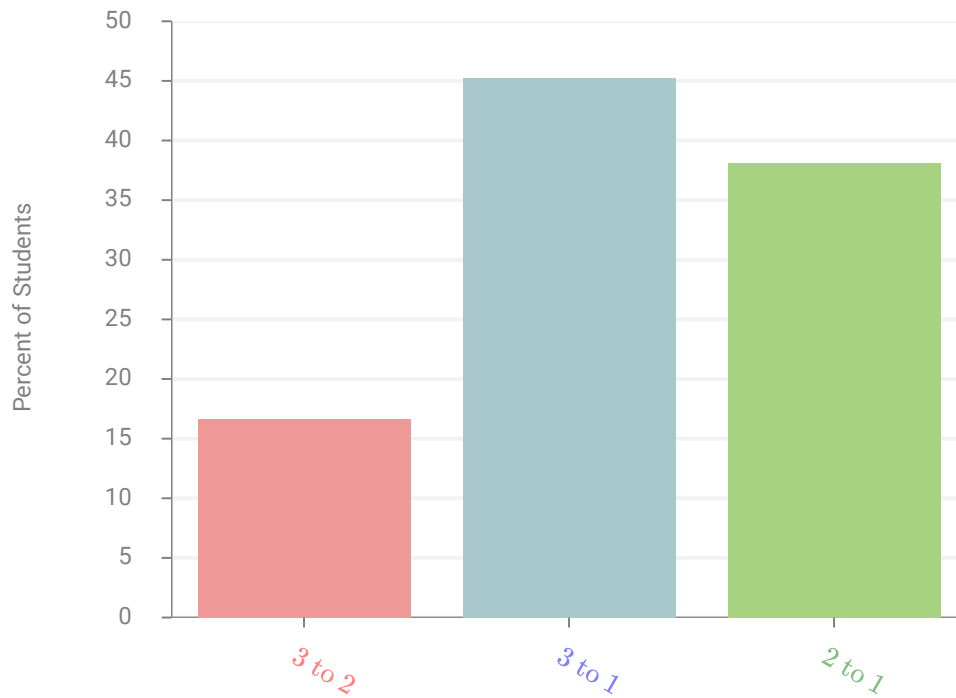
Between frame 18 and 20, McGowan traveled the distance from...

3 to 2

3 to 1

2 to 1

aninconle 2026-09-11 20:01:00



aninconle 2026-09-11 20:01:06

In frame 18, her front wheel was at line 2. In frame 20, it's at line 1. She traveled from 2 to 1.

aninconle 2026-09-11 20:01:10

Let's measure that distance using a computer tool.

aninconle 2026-09-11 20:01:12



aninconle 2026-09-11 20:01:17

Left click and drag to make a measurement from point to point. Right click to delete your last measurement.

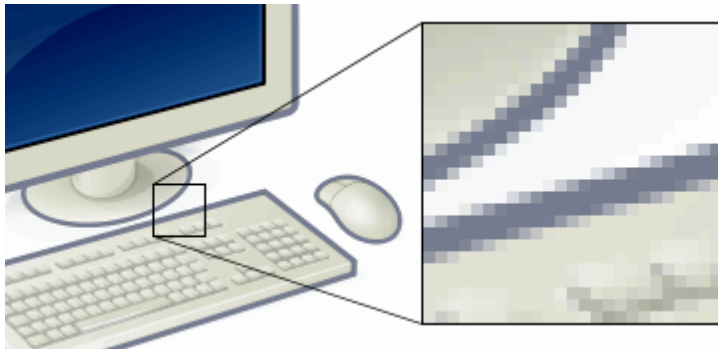
aninconle 2026-09-11 20:01:36

If you're having trouble, it's okay. I'll share screenshots so you see how it works.

aninconle 2026-09-11 20:01:40

Digital images are made of millions of small dots called *pixels*.

aninconle 2026-09-11 20:01:41



aninconle 2026-09-11 20:01:43

The tool measures distances in pixels.

aninconle 2026-09-11 20:01:49

What two things will you need to measure in pixels?

vmene 2026-09-11 20:03:29

the distance from 2 to 1 and the wheel base

hw_zasf.xAx 2026-09-11 20:03:29

size of the wheel and distance between frames

emilywow 2026-09-11 20:03:29

the wheelbase and the distance between line 1 and line 2

Oceanix 2026-09-11 20:03:29

Number of pixels covered. side length of each pixel

syl314 2026-09-11 20:03:29

The wheelbase and the distance traveled

Brightgiraffe 2026-09-11 20:03:29

the wheelbase in pixels and the distance traveled

CodeWrecker 2026-09-11 20:03:29

distance traveled & distance of wheel base

aahelic13 2026-09-11 20:03:29

the wheel base

aninconle 2026-09-11 20:03:37

We'll need the wheelbase. This will give us a conversion factor between pixels and meters. Next, we need the distance McGowan's bike moved.

aninconle 2026-09-11 20:03:38

How long is McGowan's wheelbase?

wheelbase $\approx$ pixels

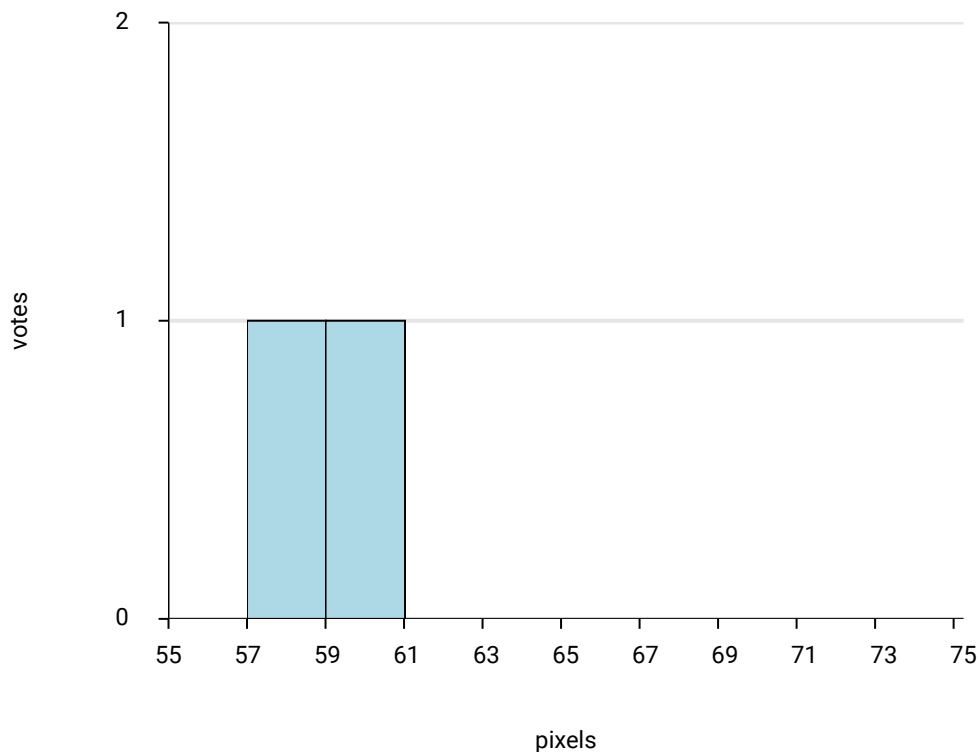
aninconle 2026-09-11 20:06:21

Our class average is 85 pixels.

aninconle 2026-09-11 20:06:51

Though, there is somethign really wrong with the histogram.

aninconle 2026-09-11 20:06:53



aninconle 2026-09-11 20:07:15

I'm assuming this is a technical issue since a lot of you put in much higher numbers haha.

Zip-a-Dee-Doo-Dah 2026-09-11 20:07:27

I put 81 in for example.

aninconle 2026-09-11 20:07:44

Yeah, like I said, the average is much higher than that shows.

aninconle 2026-09-11 20:07:53

Anyways:

aninconle 2026-09-11 20:07:54

That means

aninconle 2026-09-11 20:07:55

$$85 \text{ pixels} \approx 0.977 \text{ m}$$

aninconle 2026-09-11 20:07:59

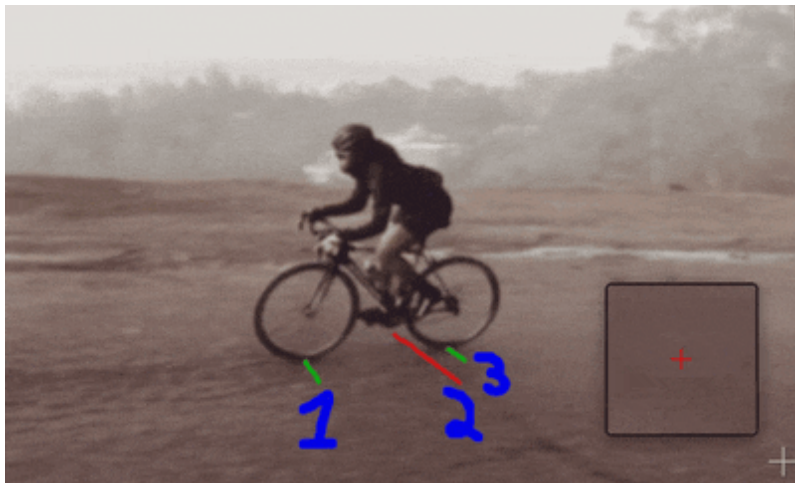
How far did McGowan move between frames 18 and 20?

distance moved $\approx$ pixels

aninconle 2026-09-11 20:10:24

Ok, the histogram software is clearly broken toady. Because it only recored 9 data points.

aninconle 2026-09-11 20:10:32

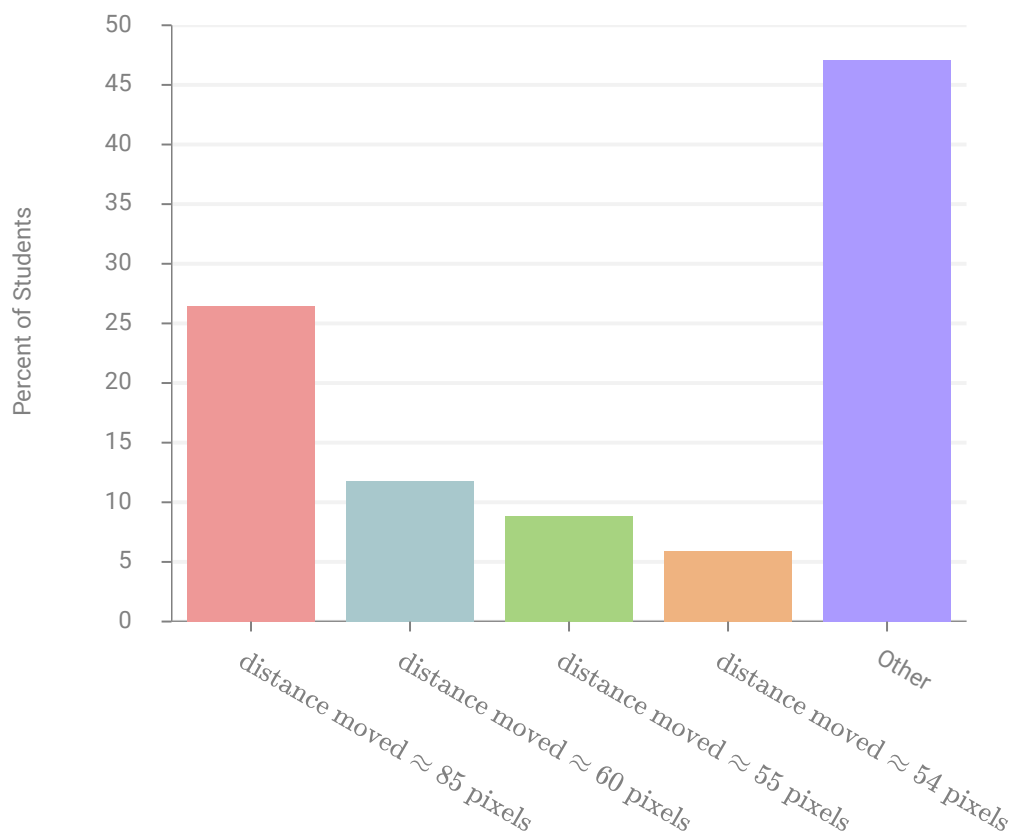


aninconle 2026-09-11 20:10:34

We averaged 63 pixels.

aninconle 2026-09-11 20:10:40

How far did McGowan move between frames 18 and 20?



aninconle 2026-09-11 20:11:02

The bar chart kind of works 🐼 But it doesn't really convey the information.

aninconle 2026-09-11 20:11:10

Anyway:

aninconle 2026-09-11 20:11:10

How do we find the distance she moved in meters?

Seifr 2026-09-11 20:13:26

By using the conversion of 0.977 m to 85 pixels.

hw_zasf.xAx 2026-09-11 20:13:26

convert pixels to a standard measurement

Marshall_Huang 2026-09-11 20:13:26

use the ratio 85 pixels = 1 meter

mocha_x 2026-09-11 20:13:26

use the bike wheelbase reference

Canada1334 2026-09-11 20:13:26

Well if we know the distance of the wheelbase is 0.977 m and the wheel base is 85 pixels, we can measure the distance travelled in pixels and convert it to meters

aninconle 2026-09-11 20:13:28

Let's use the wheelbase as a conversion factor.

$$1 = \frac{0.977 \text{ m}}{85 \text{ pixels}}$$

aninconle 2026-09-11 20:13:30

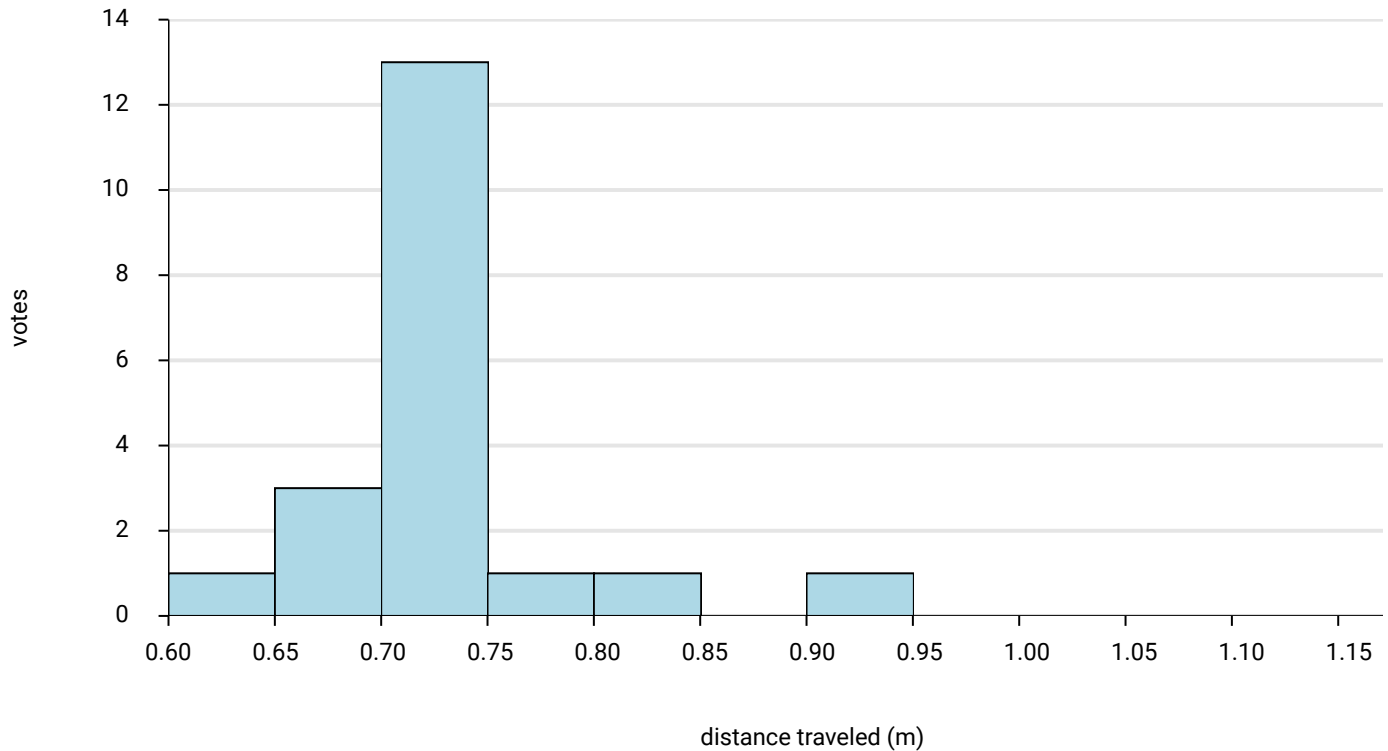
How far did McGowan travel between frames 18 and 20?

distance traveled $\approx$ m

aninconle 2026-09-11 20:14:57

Hey, this one works:

aninconle 2026-09-11 20:14:59



aninconle 2026-09-11 20:15:03

The distance was

$$\begin{aligned} \text{distance} &\approx 63 \text{ pixels} \cdot \left(\frac{0.977 \text{ m}}{85 \text{ pixels}} \right) \\ &\approx 0.72 \text{ m.} \end{aligned}$$

aninconle 2026-09-11 20:15:07

How do we get her speed?

flybirdeee_ 2026-09-11 20:15:58

divide by the time

VeryDragon2013 2026-09-11 20:15:58

divide it by the time

apollo2000 2026-09-11 20:15:58

find out the time passed between the frames and divide distance by time

kanishka.chittoor 2026-09-11 20:15:58

divide distance by time(frames

Oceanix 2026-09-11 20:15:58

find the time and divide 0.72m/time

Canada1334 2026-09-11 20:15:58

We need time

kanishka.chittoor 2026-09-11 20:15:58

divide distance by time(frames)

aahelic13 2026-09-11 20:15:58

the distance/time

aninconle 2026-09-11 20:16:00

We divide by the time it took her. Remember we used frames 18 and 20, and the frame rate is $0.05 \frac{\text{s}}{\text{frame}}$.

aninconle 2026-09-11 20:16:02

How much time passed between our two frames?

kabir_sahil 2026-09-11 20:17:06

0.1 secs

CodeWrecker 2026-09-11 20:17:06

1/10 s

Ash_Ketchup13 2026-09-11 20:17:06

0.1 s

Diamond_Zombie67 2026-09-11 20:17:06

0.1 sec

cicitheduckyyy 2026-09-11 20:17:06

1/10 of a second

AllenLuo 2026-09-11 20:17:06

0.1 seconds

emilywow 2026-09-11 20:17:06

0.1s

VeryDragon2013 2026-09-11 20:17:06

0.1 seconds

Thorongil 2026-09-11 20:17:06

0.1 s

Sefr 2026-09-11 20:17:06

0.1 seconds.

aahelic13 2026-09-11 20:17:06

0.1 seconds

Nebula233 2026-09-11 20:17:06

0.1 seconds

syl314 2026-09-11 20:17:06

1/10 s

M.Zhang555 2026-09-11 20:17:06

0.1 seconds

clairexc 2026-09-11 20:17:06

0.1 secs

Marshall_Huang 2026-09-11 20:17:06

0.1 seconds

Zip-a-Dee-Doo-Dah 2026-09-11 20:17:06

1/10th of a second.

aninconle 2026-09-11 20:17:13

Two frames passed at 0.05 s each, so that's 0.1 s .

aninconle 2026-09-11 20:17:15

distance traveled = 0.72 m

time elapsed = 0.10 s .

aninconle 2026-09-11 20:17:17

How fast was McGowan going?

McGowan's speed $\approx$ m/s

Thorongil 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

emilywow 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

guacataj 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

Supertomato1919 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

Zip-a-Dee-Doo-Dah 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

WildOctopus 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

flybirdeee_ 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

Bunnicorn 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

AN29 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

aahelic13 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

Oceanix 2026-09-11 20:18:34

McGowan's speed ≈ 7.24 m/s

agoel2010 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

TatuShark 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

CAAI 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

thekitsune 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

veryhungrypanda 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

mocha_x 2026-09-11 20:18:34

McGowan's speed ≈ 7.2 m/s

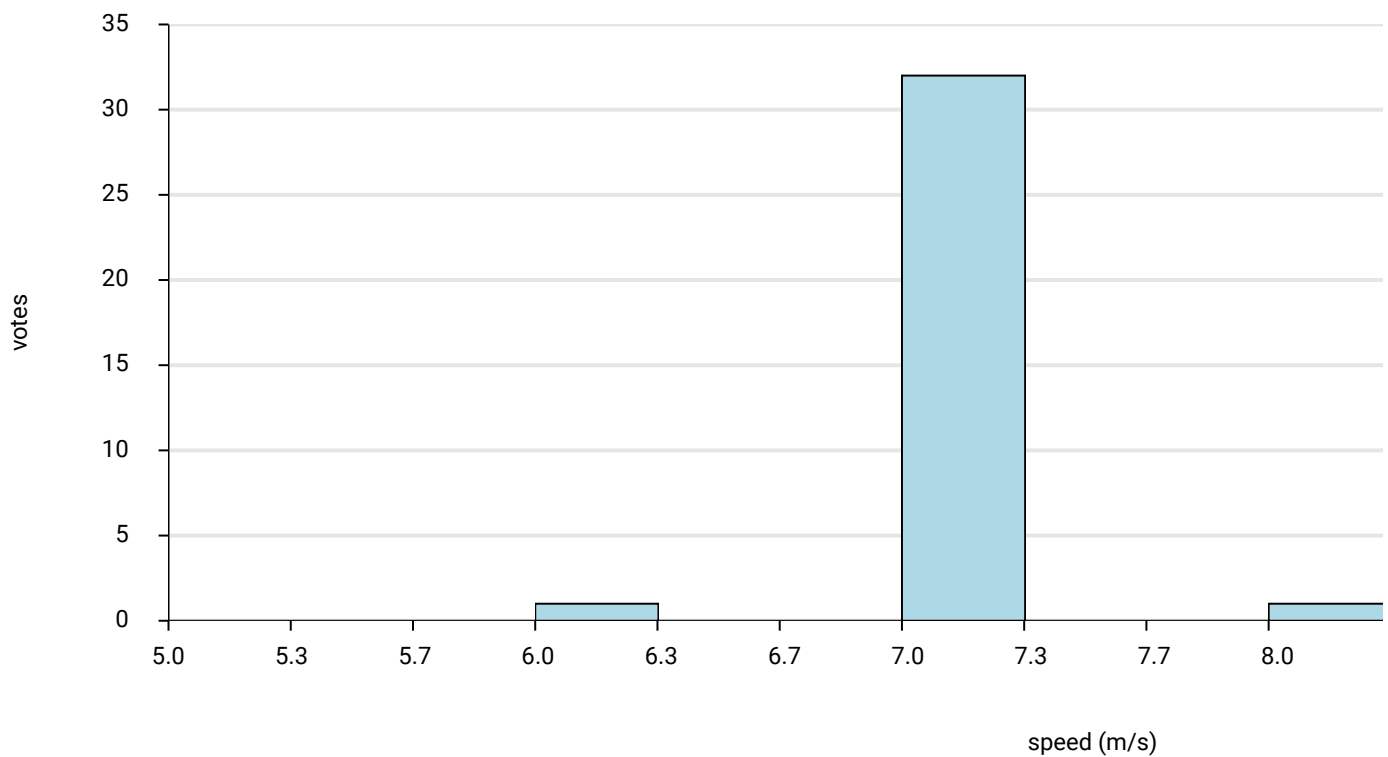
M.Zhang555 2026-09-11 20:18:34
McGowan's speed ≈ 7.2 m/s

LCLARK5691 2026-09-11 20:18:34
McGowan's speed ≈ 7.2 m/s

Nova_Wu_2013 2026-09-11 20:18:34
McGowan's speed ≈ 7.2 m/s

clairexc 2026-09-11 20:18:34
McGowan's speed ≈ 7.2 m/s

aninconle 2026-09-11 20:18:43



aninconle 2026-09-11 20:18:47

$$\begin{aligned}\text{McGowan's speed} &\approx \frac{0.72 \text{ m}}{0.10 \text{ s}} \\ &= 7.2 \text{ m/s.}\end{aligned}$$

aninconle 2026-09-11 20:18:52

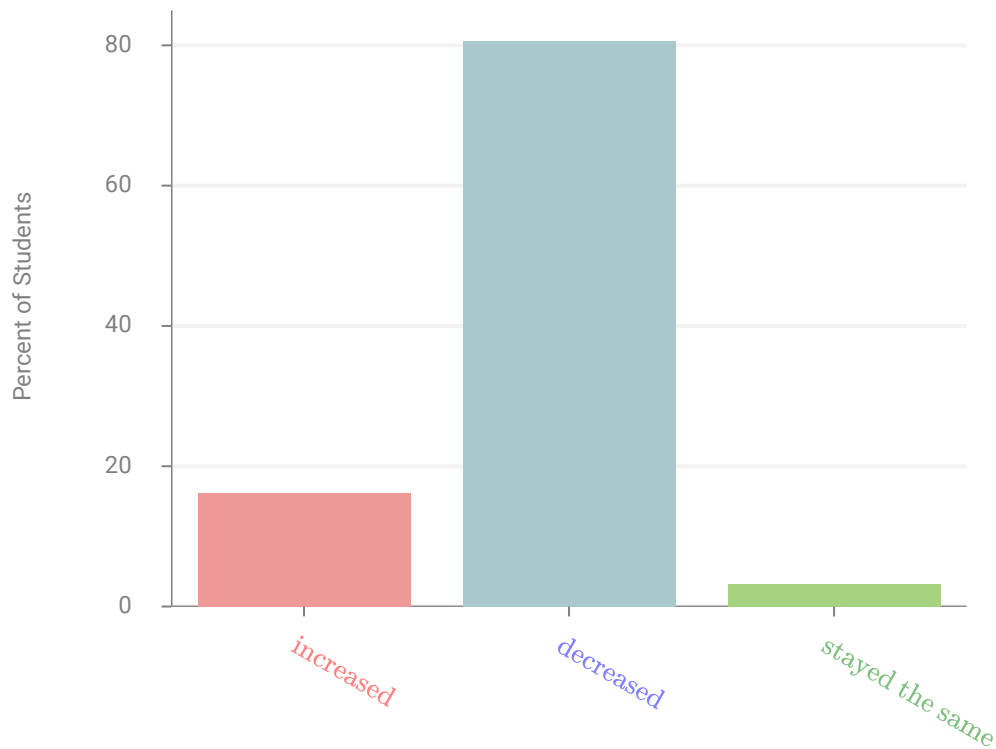
Since our first guess, our *uncertainty* has...

increased

decreased

stayed the same

aninconle 2026-09-11 20:19:38



aninconle 2026-09-11 20:19:43

It's decreased. Our estimate isn't perfect, but it's much better than guessing, like we did at first.

aninconle 2026-09-11 20:19:45

Let's estimate our uncertainty. We measured about 63 pixels for how far McGowan's bike moved, but let's say we could have been off by up to 2 pixels.

aninconle 2026-09-11 20:19:47

What's that give for our percentage uncertainty?

aahelic13 2026-09-11 20:21:05

3%

VeryDragon2013 2026-09-11 20:21:05

like 3%?

guacataj 2026-09-11 20:21:05

+ 3%

clairexc 2026-09-11 20:21:05

+ 3%

AN29 2026-09-11 20:21:05

about 3%

Oceanix 2026-09-11 20:21:05

3%

Thorongil 2026-09-11 20:21:05

$\approx 3.17\%$

Nebula233 2026-09-11 20:21:05

3%?

CodeWrecker 2026-09-11 20:21:05

3%

thekitsune 2026-09-11 20:21:05

3%

flybirdeee_ 2026-09-11 20:21:05

3% error??

VeryDragon2013 2026-09-11 20:21:05

 $\approx 3\%$

Exponent11 2026-09-11 20:21:05

3%

aninconle 2026-09-11 20:21:08

It's $\frac{2 \text{ pixels}}{63 \text{ pixels}} \approx 3\%$.

aninconle 2026-09-11 20:21:10

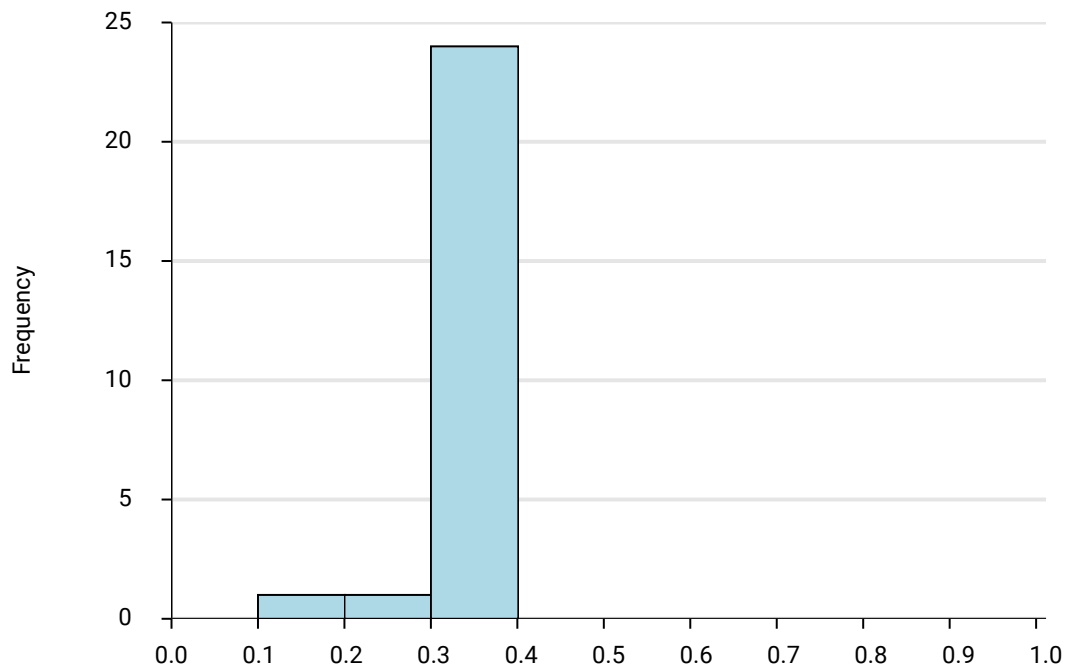
We also have some uncertainty about the wheelbase, and some about the time, so let's round up to 5% uncertainty.

aninconle 2026-09-11 20:21:11

If our final estimate of speed had 5% uncertainty, how could we write our speed estimate?

McGowan's speed = $7.0 \pm$ n/s

aninconle 2026-09-11 20:22:45



aninconle 2026-09-11 20:22:46

5% of 7.2 m/s is $(7.2 \cdot 0.05) \text{ m/s} = 0.36 \text{ m/s}$. **We round to 0.4 m/s and write**

$$\text{McGowan's speed} = 7.2 \pm 0.4 \text{ m/s}.$$

aninconle 2026-09-11 20:22:48

(Scientists normally use an equal sign when they write a quantity with an uncertainty. Even though 7.2 m/s is still an approximation, the $\pm 0.4 \text{ m/s}$ tells us more about the uncertainty than an $\approx$ sign would, and it's simply common practice to use $=$ instead.)

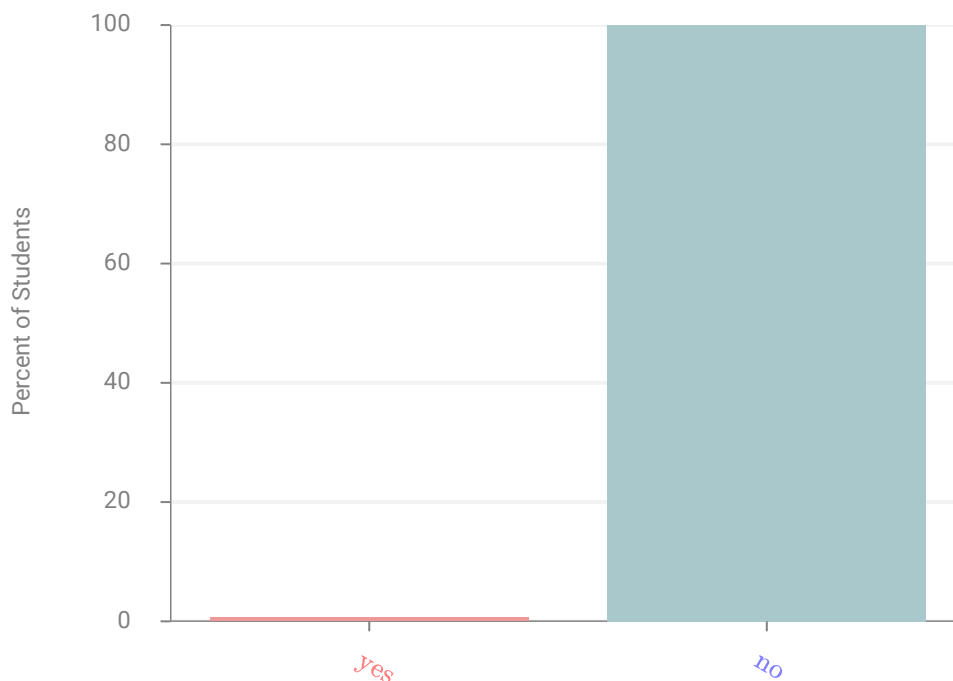
aninconle 2026-09-11 20:22:52

Do we know McGowan's speed was 7.2 m/s over the entire clip?

yes

no

aninconle 2026-09-11 20:23:33



aninconle 2026-09-11 20:23:37

No, we don't. Her speed may have varied. All we've found is her average speed over one short part!

aninconle 2026-09-11 20:23:39

Takeaway: we can make quantitative measurements of phenomena. Those may be estimates, but as we improve our techniques, we reduce our uncertainty.

aninconle 2026-09-11 20:23:40

Emilie the Skater: notation and tools for building models

aninconle 2026-09-11 20:23:44

Throughout the course, we'll continually build new conceptual models. The more language we have to do that, the more powerful those models can be.

aninconle 2026-09-11 20:23:45

Learn the basic language of motion in physics — instantaneous and average speed, dimensions, coordinates, position, displacement, velocity, Δ , positive and negative motion.

aninconle 2026-09-11 20:23:51

Physicists use some words differently than their everyday meaning. For example, *speed* and *velocity* are different in physics.

aninconle 2026-09-11 20:23:52

Our goal here is to learn new words and concepts related to measure motion.

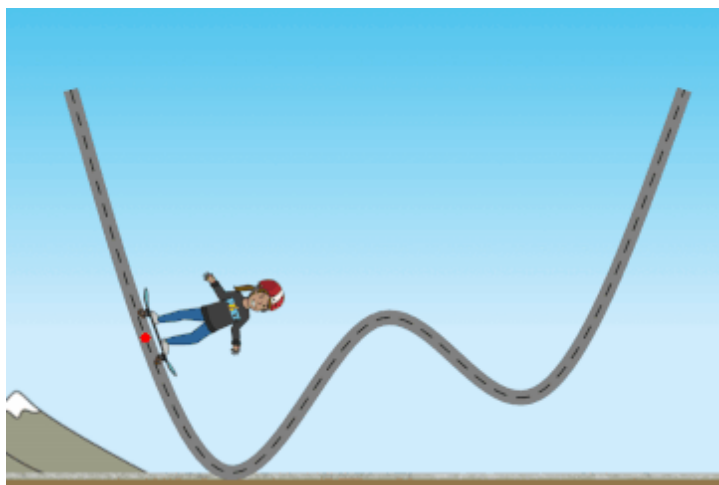
aninconle 2026-09-11 20:23:54

We'll study a skater named Emilie, in honor of Émilie du Châtelet, the French mathematician and scientist who gave mechanics the concept of energy conservation. (We'll study that later in the course.)

aninconle 2026-09-11 20:23:57

Here's Emilie skating back and forth.

aninconle 2026-09-11 20:23:58



aninconle 2026-09-11 20:24:07

In general, where is she when her speed is high?

emilywow 2026-09-11 20:25:31

at the lower points

ndmath324458 2026-09-11 20:25:31

the lowest point of the path is there her speed is the highest

M.Zhang555 2026-09-11 20:25:31

lowest point?

CodeWrecker 2026-09-11 20:25:31

on the lower parts of the track

Ash_Ketchup13 2026-09-11 20:25:31

low to tge ground

thekitsune 2026-09-11 20:25:31

the low points

TatuShark 2026-09-11 20:25:31

at the bottom

flybirdeee_ 2026-09-11 20:25:31

lower areas

syl314 2026-09-11 20:25:31

At the bottom of the big downhill

Seifr 2026-09-11 20:25:31

The lowest part.

LCLARK5691 2026-09-11 20:25:31
at the bottom of the first downhill

agoel2010 2026-09-11 20:25:31
when she first drops so the highest points

Bunnicorn 2026-09-11 20:25:31
at the bottom

aninconle 2026-09-11 20:25:36

Emilie goes faster the lower down she is. That makes sense – when Emilie goes downhill, gravity speeds her up.

aninconle 2026-09-11 20:25:39



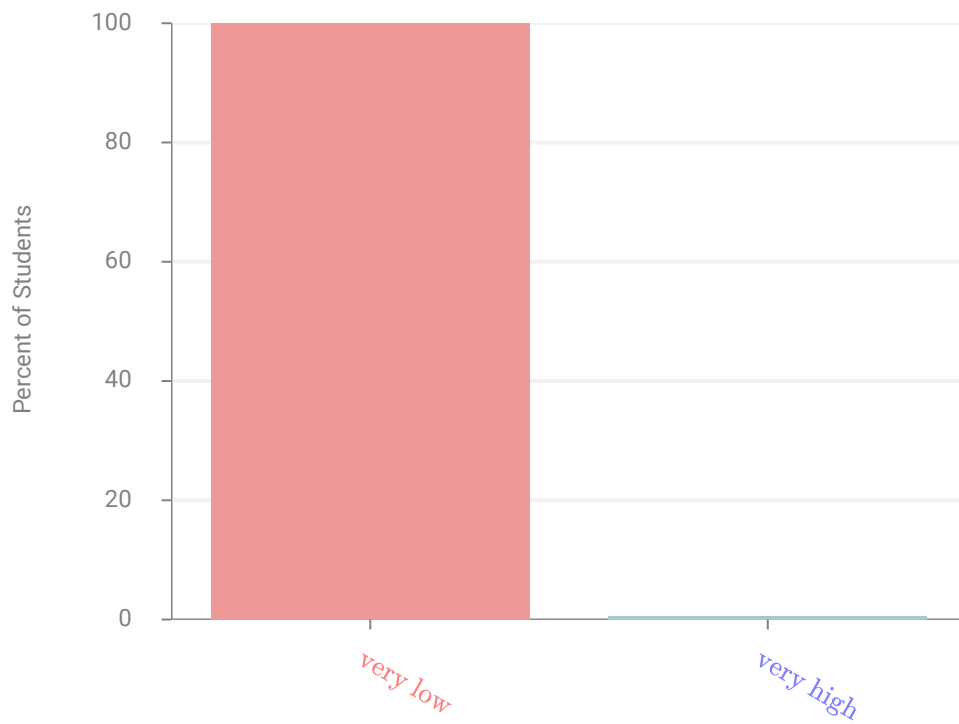
aninconle 2026-09-11 20:25:47

When Emilie nears her highest point, on either side, her speed gets...

very low

very high

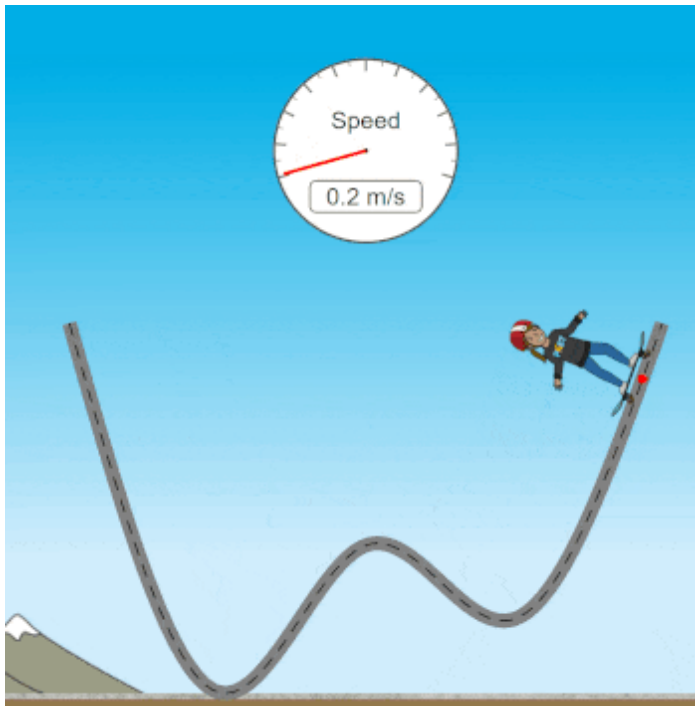
aninconle 2026-09-11 20:26:37



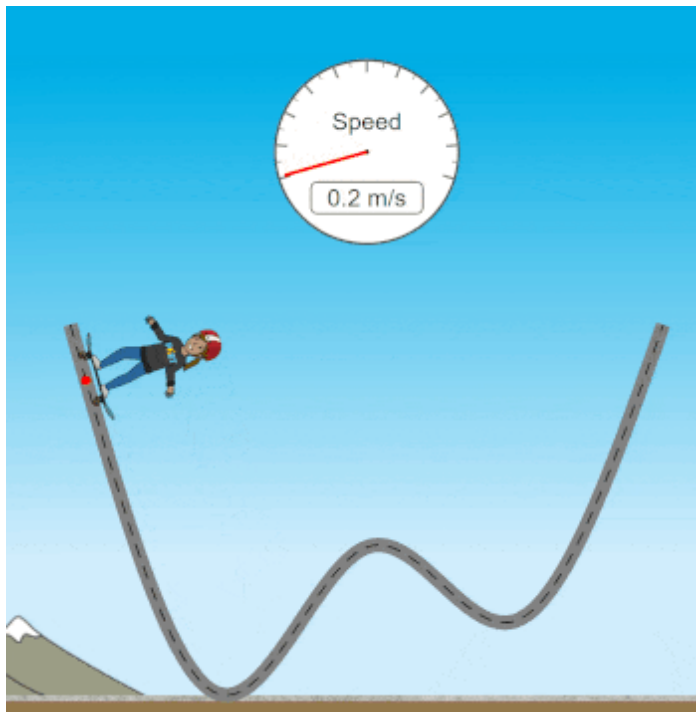
aninconle 2026-09-11 20:26:44

It gets very low, since it drops to zero right when she turns around.

aninconle 2026-09-11 20:26:45



aninconle 2026-09-11 20:26:46



aninconle 2026-09-11 20:26:48

Emilie's speed is always changing. That makes it harder to measure.

aninconle 2026-09-11 20:26:49

For example, the simulation has a "path" feature. It leaves a trail of purple dots behind Emilie.

aninconle 2026-09-11 20:26:50



aninconle 2026-09-11 20:26:53

The simulation draws one dot every 0.1 s. The result is called a *dot diagram*.

aninconle 2026-09-11 20:26:56

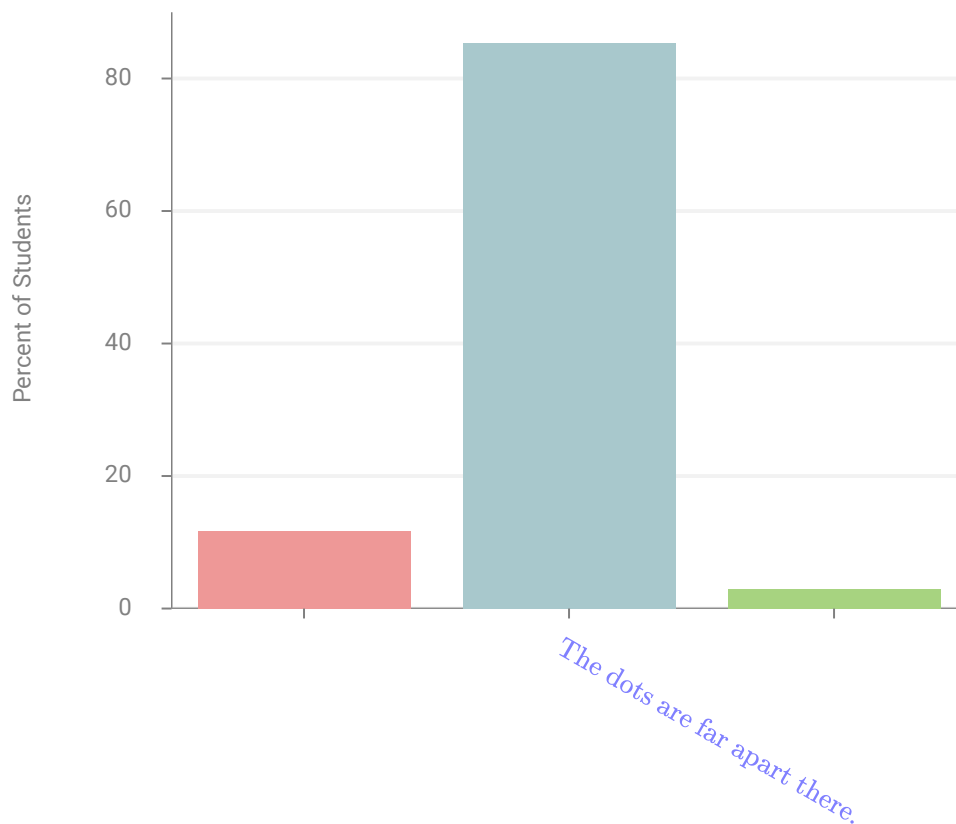
If you only saw the dots, how would you tell where Emilie was going fast?

The dots are close together there.

The dots are far apart there.

You can't tell from the dots.

aninconle 2026-09-11 20:27:36



aninconle 2026-09-11 20:27:39

If Emilie is going fast, she moves a long distance in 0.1 s, so the dots will be far apart.

aninconle 2026-09-11 20:27:41

We could divide the distance between two dots by 0.1 s to get Emilie's speed. Why won't that give us her exact speed at the moment the first dot was laid down?

VeryDragon2013 2026-09-11 20:28:58

Because it's changing and not constant

Diamond_Zombie67 2026-09-11 20:28:58

because the speed is changing

vmene 2026-09-11 20:28:58

because it is the average speed between those 2 points, it won't that give us her exact speed at the moment the first dot was laid down

flybirdeee_ 2026-09-11 20:28:58

the speed changes

hw_zasf.xAx 2026-09-11 20:28:58

because her speed changes

kanishka.chittoor 2026-09-11 20:28:58

she is constantly accelerating

Thorongil 2026-09-11 20:28:58

that would be the average speed over that interval, not the instantaneous speed

Marshall_Huang 2026-09-11 20:28:58

that is just the average, speed changing

Nebula233 2026-09-11 20:28:58

her speed changes

apollo2000 2026-09-11 20:28:58

She might have accelerated after then.

aninconle 2026-09-11 20:29:00

Emilie's speed changes, even in 0.1 s. We would find her *average* speed over 0.1 s this way.

aninconle 2026-09-11 20:29:01

We want Emilie's *instantaneous* speed. To describe that, let's develop new language about motion.

aninconle 2026-09-11 20:29:05

First, let's describe Emilie's *position* — where she is on her track.

aninconle 2026-09-11 20:29:06



aninconle 2026-09-11 20:29:11

What feature on the drawing could we count as a measurement of position?

aninconle 2026-09-11 20:30:03

(Hint: It's not the dots -- those are changing as Emilie's speed is changing, so they're not a good marker of position.)

syl314 2026-09-11 20:30:33

That squiggly line?

raphaelyuan 2026-09-11 20:30:33

distance along the path

AN29 2026-09-11 20:30:33

the lines

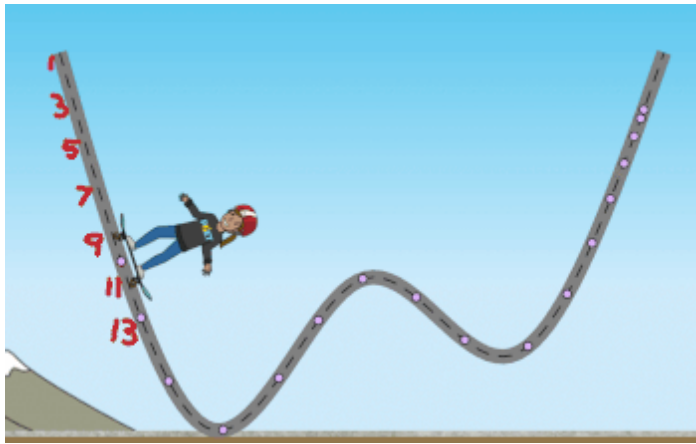
emilywow 2026-09-11 20:30:33

maybe those black lines

aninconle 2026-09-11 20:30:34

We count the dashes, starting from the left.

aninconle 2026-09-11 20:30:36



aninconle 2026-09-11 20:30:40

Emilie is at dash 10. There are four dashes per meter in this simulation.

aninconle 2026-09-11 20:30:41

How far is Emilie from the left edge of the track, measured along the track?

 m

Sefr 2026-09-11 20:32:13

2.5 m

M.Zhang555 2026-09-11 20:32:13

2.5 m

kabir_sahil 2026-09-11 20:32:13

2.5 m

apollo2000 2026-09-11 20:32:13

2.5 m

syl314 2026-09-11 20:32:13

2.5 m

WildOctopus 2026-09-11 20:32:13

2.5 m

rysaishi 2026-09-11 20:32:13

2.5 m

Zip-a-Dee-Doo-Dah 2026-09-11 20:32:13

2.5 m

thekitsune 2026-09-11 20:32:13

2.5 m

AllenLuo 2026-09-11 20:32:13

2.5 m

kanishka.chittoor 2026-09-11 20:32:13

2.5 m

Canada1334 2026-09-11 20:32:13

2.5 m

Marshall_Huang 2026-09-11 20:32:13

2.5 m

CodeWrecker 2026-09-11 20:32:13

2.5 m

Potatomiaomi 2026-09-11 20:32:13

2.5 m

TatuShark 2026-09-11 20:32:13

2.5 m

Oceanix 2026-09-11 20:32:13

2.5 m

Bunnicorn 2026-09-11 20:32:13

2.5 m

aahelic13 2026-09-11 20:32:13

2.5 m

VeryDragon2013 2026-09-11 20:32:13

2.5 m

Ash_Ketchup13 2026-09-11 20:32:13

2.5 m

guacataj 2026-09-11 20:32:13

2.5 m

CAAI 2026-09-11 20:32:13

2.5 m

aninconle 2026-09-11 20:32:16

Emilie is $\frac{10 \text{ dashes}}{4 \text{ dashes/m}} = 2.5 \text{ m}$ from the left edge of the track. We could write

$$x = 2.5 \text{ m.}$$

aninconle 2026-09-11 20:32:18

When we use a number to describe Emilie's position, that number is a *coordinate*. Here, x is our variable for Emilie's coordinate.

aninconle 2026-09-11 20:32:20

Since we only need one number to track Emilie's position, her motion is *one-dimensional*. Your computer screen is two-dimensional, but Emilie is stuck on a one-dimensional (though bent) track.

aninconle 2026-09-11 20:32:23

We now have a *coordinate system*. x is a *coordinate* that tells us where Emilie is.

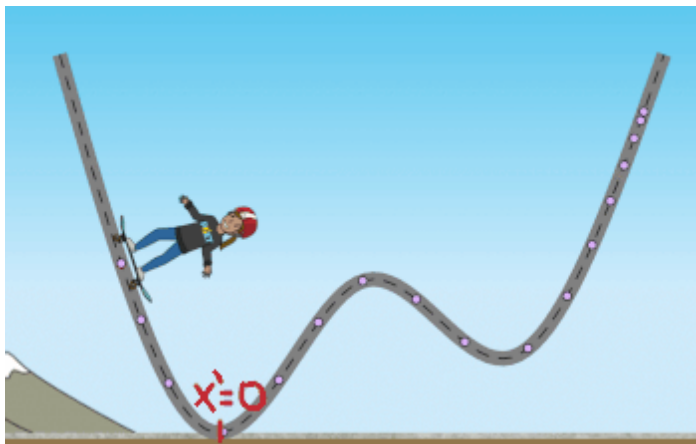
aninconle 2026-09-11 20:32:24

Emilie's *position* is slightly different than her coordinate. Her position is the actual place that she is — a specific spot on the ramp. Her coordinate is the number we assign to that place.

aninconle 2026-09-11 20:32:26

We set $x = 0$ at the left end of the ramp, but we could set the zero coordinate at the lowest point of the ramp, like this:

aninconle 2026-09-11 20:32:28



aninconle 2026-09-11 20:32:31

We've put a ' on x' to indicate this is a new coordinate system, different from x .

aninconle 2026-09-11 20:32:36

The pronunciation of x' is "x prime."

aninconle 2026-09-11 20:32:38

When we define a new coordinate system, what does that change?

Emilie's position

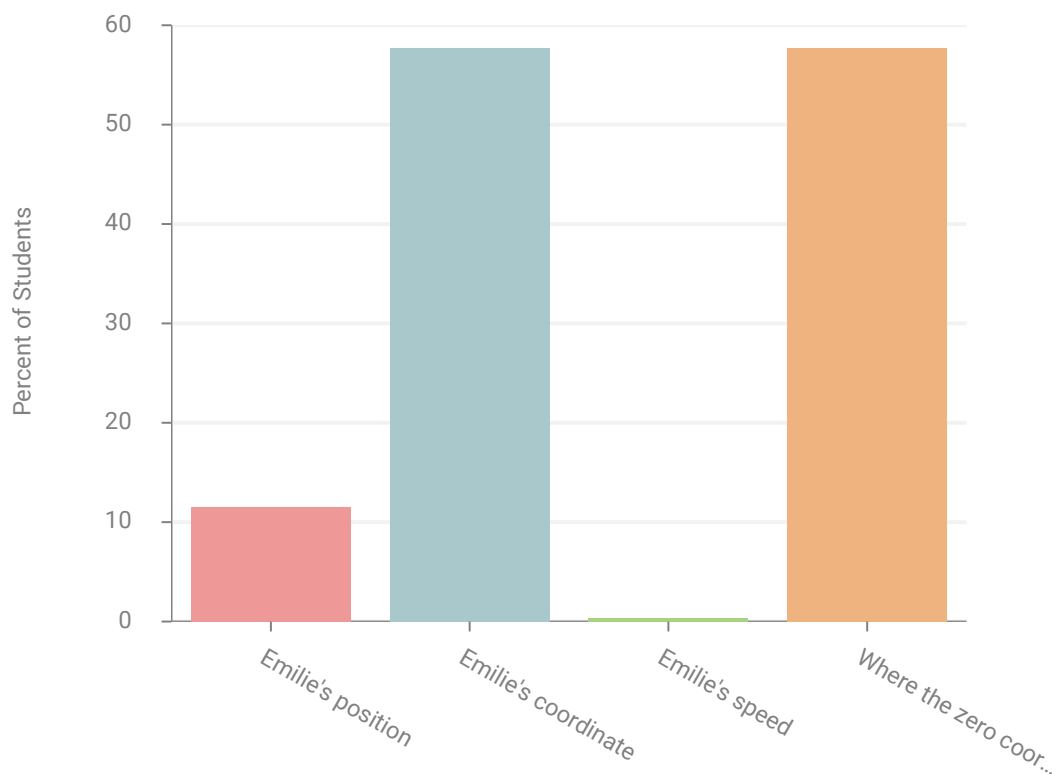
Emilie's coordinate

Emilie's speed

Where the zero coordinate is

aninconle 2026-09-11 20:33:20

When we define a new coordinate system, what does that change?



aninconle 2026-09-11 20:33:25

Emilie's position doesn't change. She's over the same part of the ramp as before. Her speed doesn't change, either. The location of zero does change, and so does Emilie's coordinate.

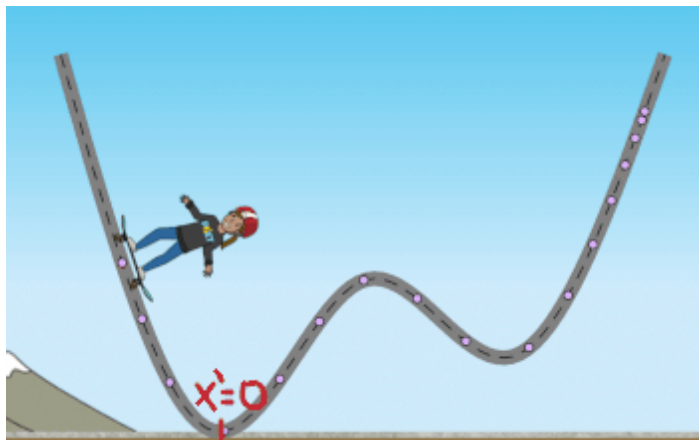
aninconle 2026-09-11 20:33:29

You may hear things like "Emilie's position is $x = 2.5$ m." This mixes up position and coordinate — Emilie's *coordinate* is $x = 2.5$ m and her *position* is whatever part of the ramp she's at. However, these sorts of minor imprecise statements are so common we have to get used to them.

aninconle 2026-09-11 20:33:32

Here's Emilie again.

aninconle 2026-09-11 20:33:33



aninconle 2026-09-11 20:33:38

She's left of $x' = 0$.

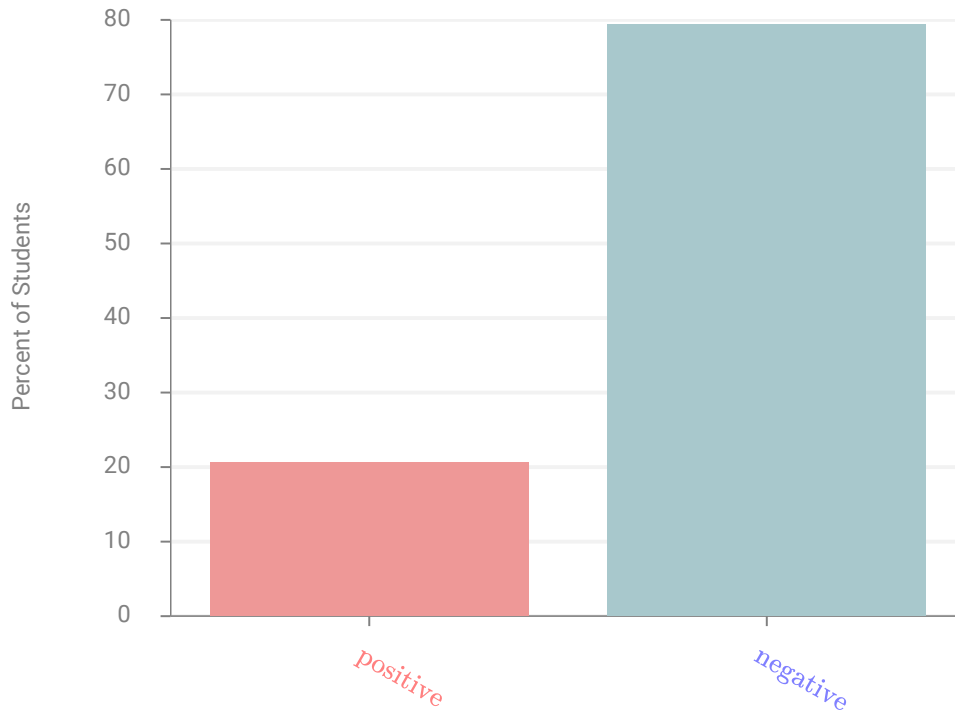
aninconle 2026-09-11 20:33:40

What type of number should we use for her coordinate?

positive

negative

aninconle 2026-09-11 20:34:52



aninconle 2026-09-11 20:34:56

It's convention for coordinates to decrease to the left. That makes Emilie's x' coordinate *negative*.

aninconle 2026-09-11 20:34:59

Motion can also be positive or negative. What does it mean if Emilie's *motion* is negative?

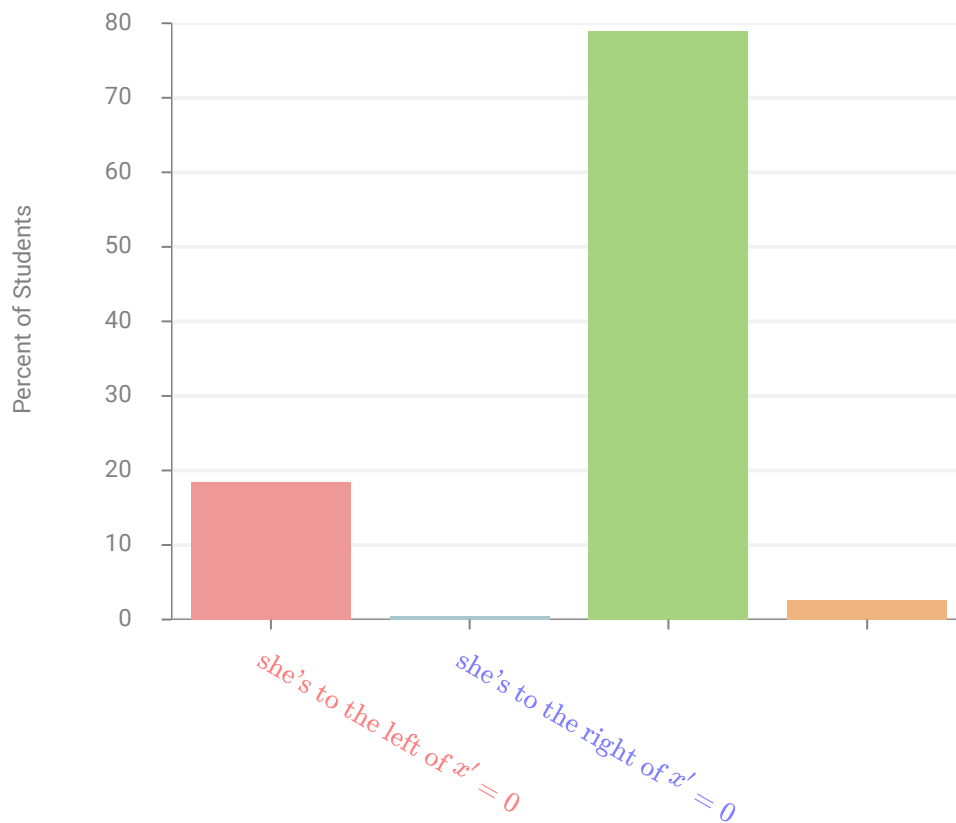
she's to the left of $x' = 0$

she's to the right of $x' = 0$

she's moving to the left (wherever she is)

she's moving to the right (wherever she is)

aninconle 2026-09-11 20:35:58



aninconle 2026-09-11 20:36:05

That means Emilie is moving left. Her x (and x') coordinate is decreasing. Her position is irrelevant to whether her motion is positive or negative. It only matters what direction she's going.

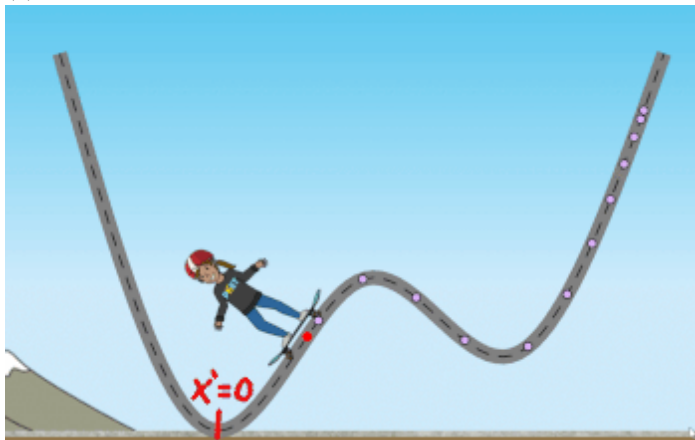
aninconle 2026-09-11 20:36:07

You can tell which direction Emilie is moving, even in a still image, because she always faces that way. In the picture above, she's moving to the left.

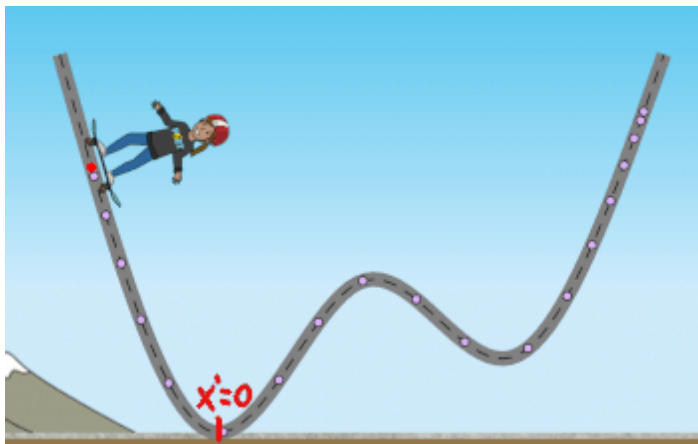
aninconle 2026-09-11 20:36:09

Select all the images where Emilie has negative motion.

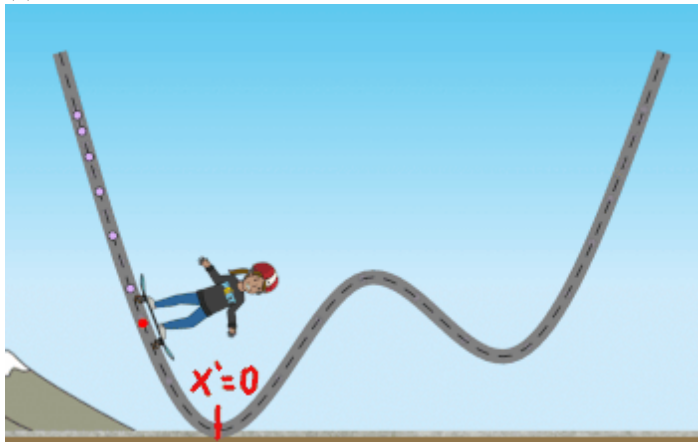
(a)



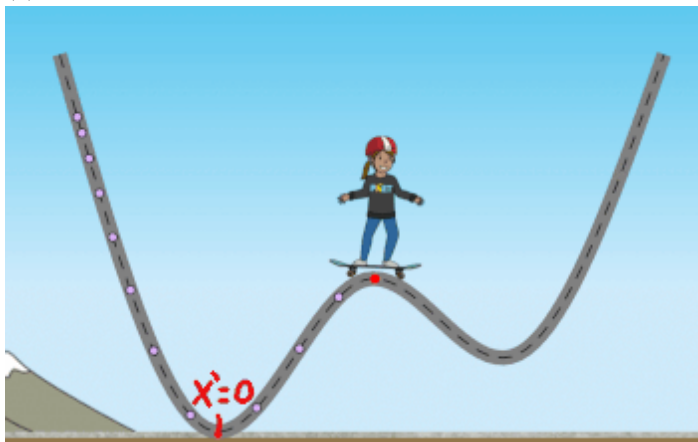
(b)



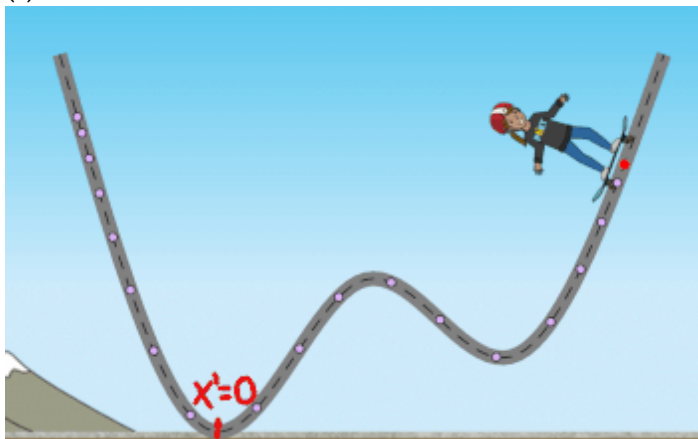
(c)



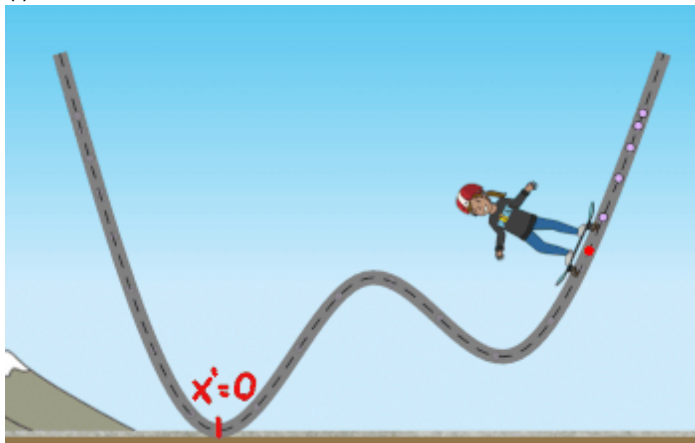
(d)



(e)



(f)



(a)

(b)

(c)

(d)

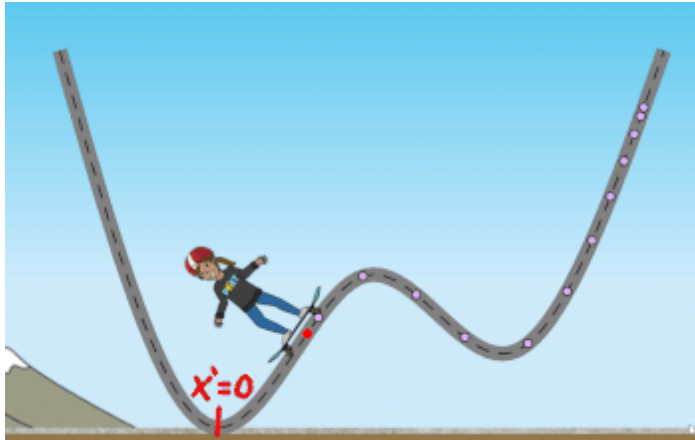
(e)

(f)

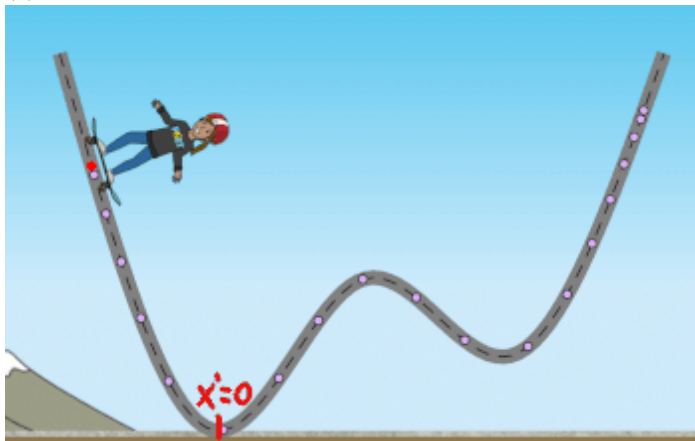
aninconle 2026-09-11 20:36:59

Select all the images where Emilie has negative motion.

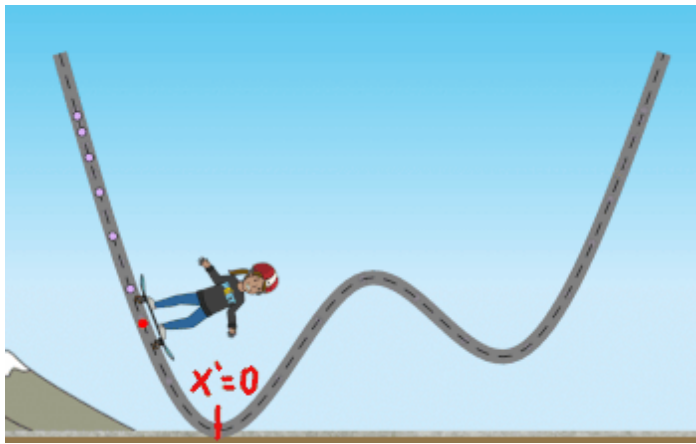
(a)



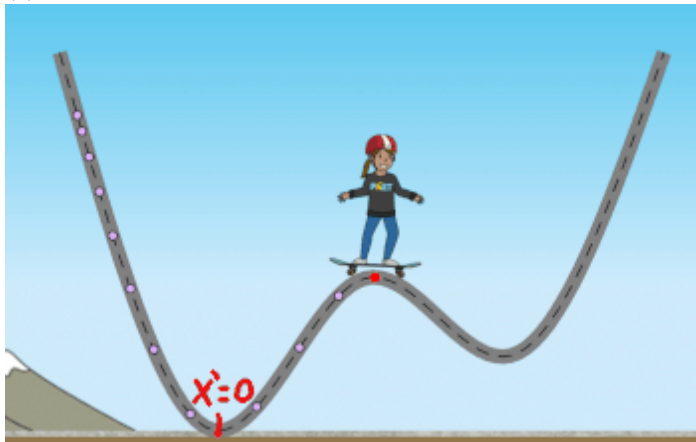
(b)



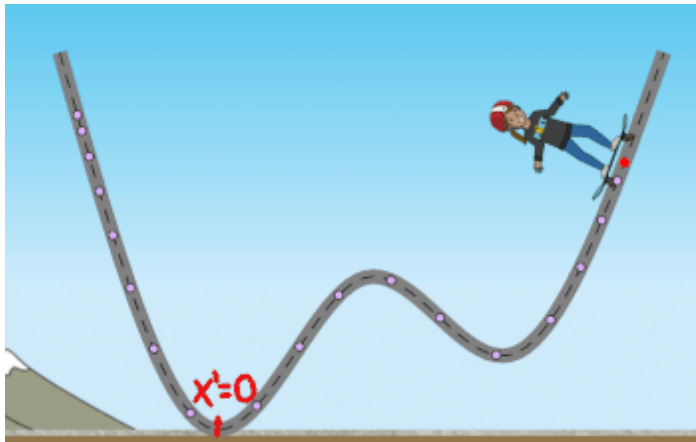
(c)



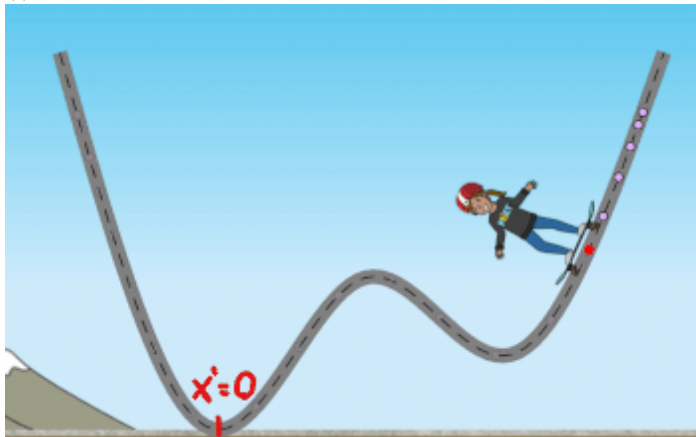
(d)

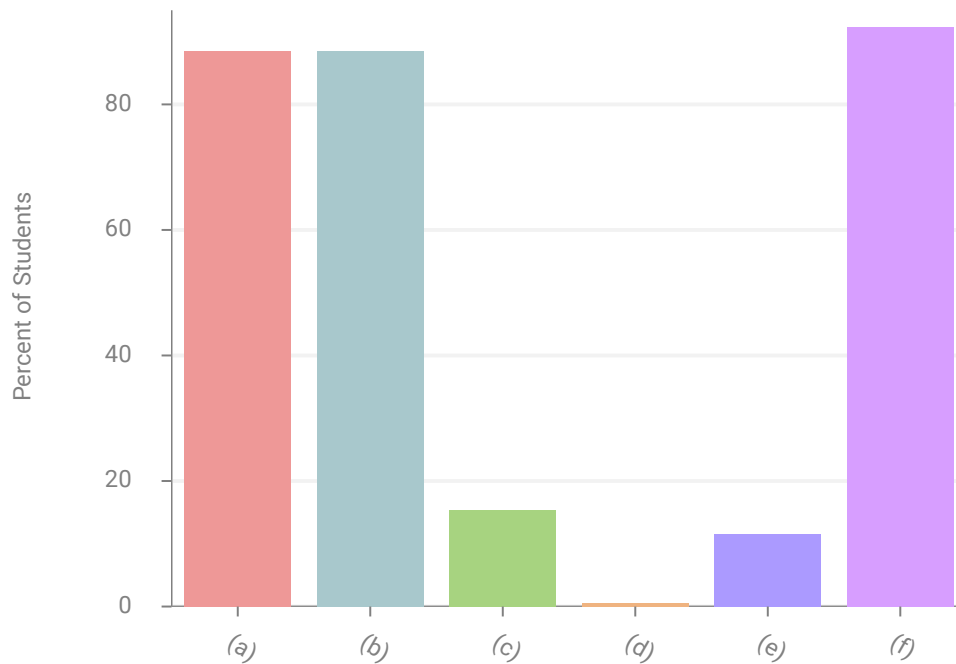


(e)



(f)



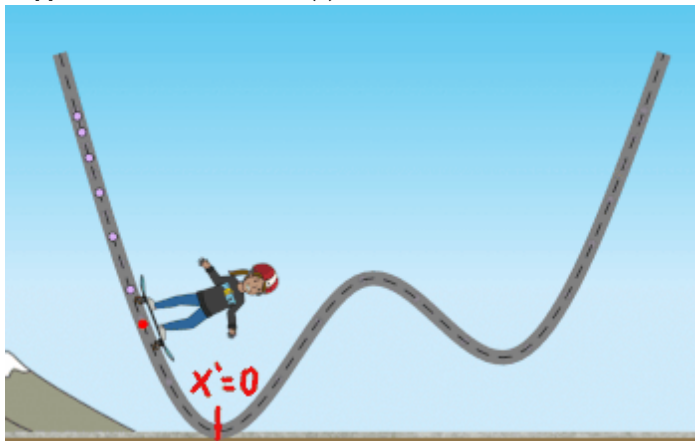


aninconle 2026-09-11 20:37:06

It's when she's headed left. The correct answers are (a), (b), and (f), since those are the ones where she's facing to the left.

aninconle 2026-09-11 20:37:07

Suppose someone selected (c).



What were they probably thinking?

flybirdeee_ 2026-09-11 20:38:33

on the left

apollo2000 2026-09-11 20:38:33

She is on the left of the 0 coordinate, so she must be negatively moving

LCLARK5691 2026-09-11 20:38:33

to the left of $x=0$

hw_zasf.xAx 2026-09-11 20:38:33

behind prime x is always negative

raphaelyuan 2026-09-11 20:38:33

That the x coordinate is negative

Sefr 2026-09-11 20:38:33

She has negative coordinates.

Canada1334 2026-09-11 20:38:33

she is left of x'

WildOctopus 2026-09-11 20:38:33

they had negative coordinate

kanishka.chittoor 2026-09-11 20:38:33

she is on left so she negative

clairexc 2026-09-11 20:38:33

She is on the left side, so her motion is negative

Oceanix 2026-09-11 20:38:33

they might be thinking "well maybe since shes going down its negative motion"

TatuShark 2026-09-11 20:38:33

That she's on the left of x' so her motion is negative

Bunnicorn 2026-09-11 20:38:33

she is on the left of x'

kabir_sahil 2026-09-11 20:38:33

that she was at a negative position so her motion would be negative

flybirdeee_ 2026-09-11 20:38:33

it was on a negative slope

syl314 2026-09-11 20:38:33

Negative coordinate, not negative motion

Zip-a-Dee-Doo-Dah 2026-09-11 20:38:33

She is on the **left** side of the point $x' = 0$

Brightgiraffe 2026-09-11 20:38:33

the coordinate is negative so the motion must be negative

Thorongil 2026-09-11 20:38:44

she's to the left of x' , so if you mix up motion with coordinate, you would think (c) is right

aninconle 2026-09-11 20:38:46

They probably noticed that Emilie's *coordinate* is negative. However, the coordinate is negative *and increasing* because Emilie is going right. Motion is about the *change* in Emilie's coordinate, not the actual value of the coordinate.

aninconle 2026-09-11 20:38:48

Sometimes we only care about how fast something is going. For example, in a car crash test it only matters how fast the car is going when it hits a wall, not whether it was going left or right.

aninconle 2026-09-11 20:38:49

When we don't care about the direction, we talk about an object's *speed*.

aninconle 2026-09-11 20:38:51

Other times, the direction is important. For example, Emilie needs to make sure she's looking in the direction she's going, so direction matters. When we care about both how fast something is moving and the direction it's moving, we use the *velocity*.

aninconle 2026-09-11 20:38:56

Velocity is positive for motion to the right and negative for motion to the left. Speed is always positive (or zero).

aninconle 2026-09-11 20:39:09

Speed is the absolute value of velocity. In physics, we often replace the word "absolute value" with *magnitude*, so speed is the magnitude of velocity.

aninconle 2026-09-11 20:39:12

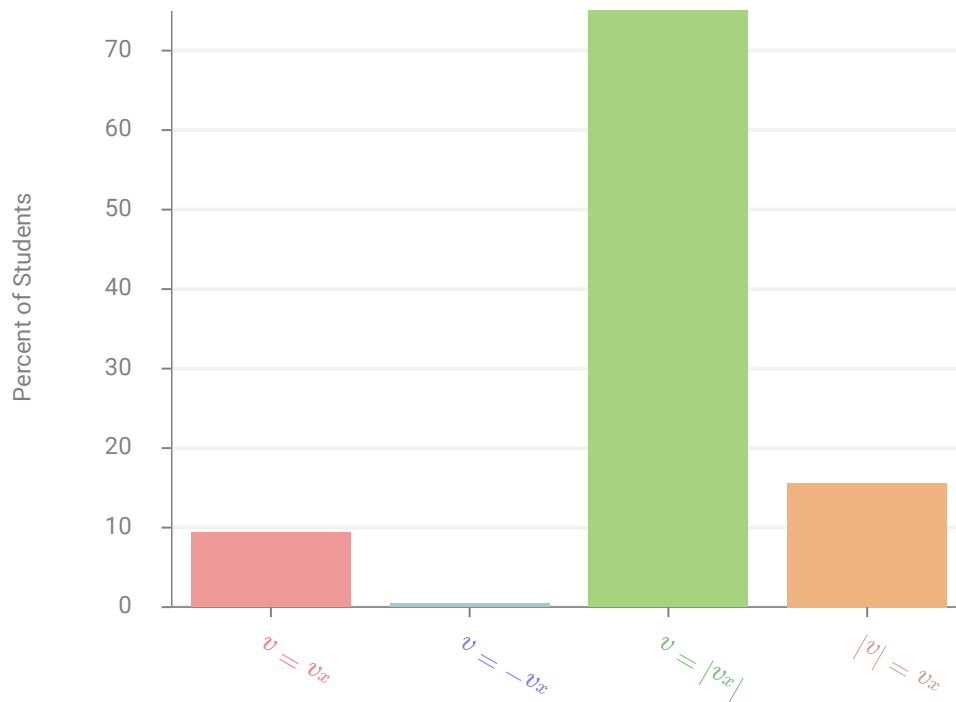
Our symbol for velocity is v_x . The subscript reminds us that v_x is the rate of change of x .

aninconle 2026-09-11 20:39:16

Our symbol for speed is v , so no matter which direction Emilie goes ...

$$v = v_x \quad v = -v_x \quad v = |v_x| \quad |v| = v_x$$

aninconle 2026-09-11 20:40:49



aninconle 2026-09-11 20:40:57

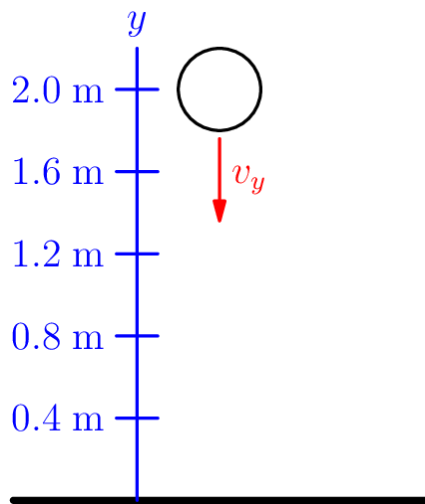
Speed is the magnitude of velocity,

$$v = |v_x|.$$

aninconle 2026-09-11 20:40:59

In other scenarios, we might use y as a coordinate, for example for height of a ball above the ground. Then as the ball fell, its velocity would be v_y .

aninconle 2026-09-11 20:41:00

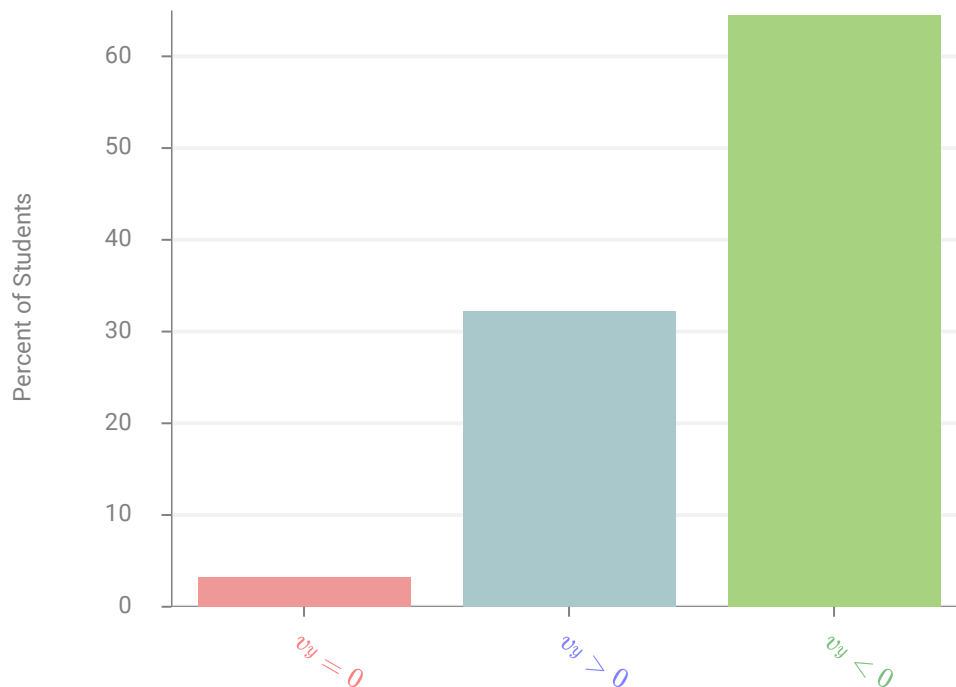


aninconle 2026-09-11 20:41:02

For the ball, ...



aninconle 2026-09-11 20:42:06



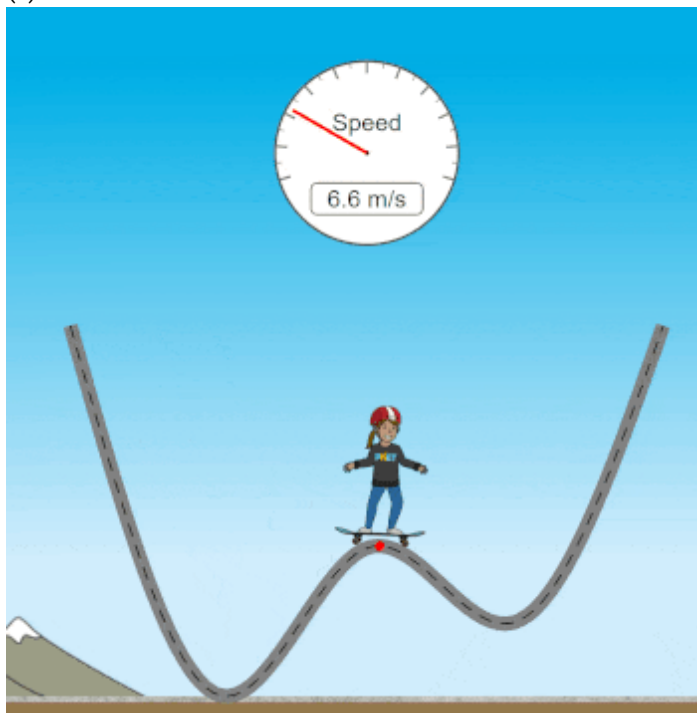
aninconle 2026-09-11 20:42:10

It's $v_y < 0$, since the y coordinate is decreasing.

aninconle 2026-09-11 20:42:16

Select two images where Emilie has the same speed, but different velocities.

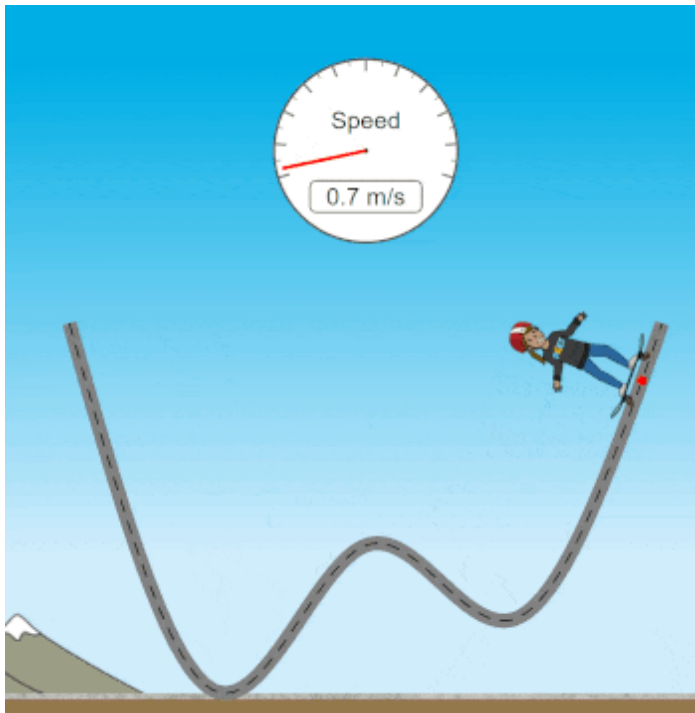
(a)



(b)



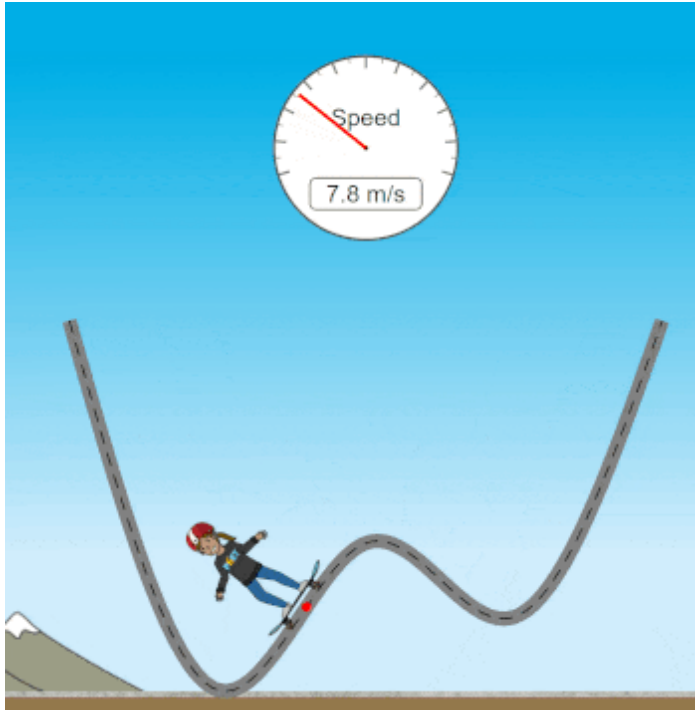
(c)



(d)



(e)



(a)

(b)

(c)

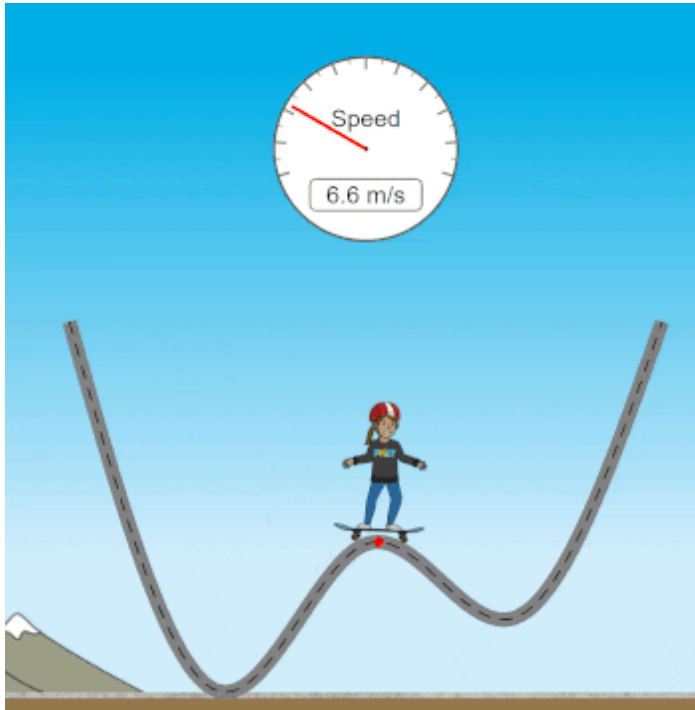
(d)

(e)

aninconle 2026-09-11 20:43:34

Select two images where Emilie has the same speed, but different velocities.

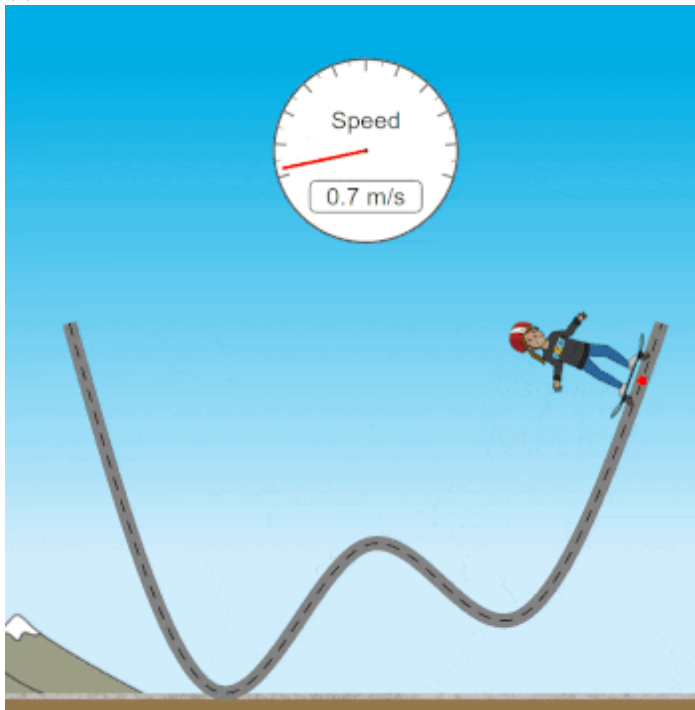
(a)



(b)



(c)

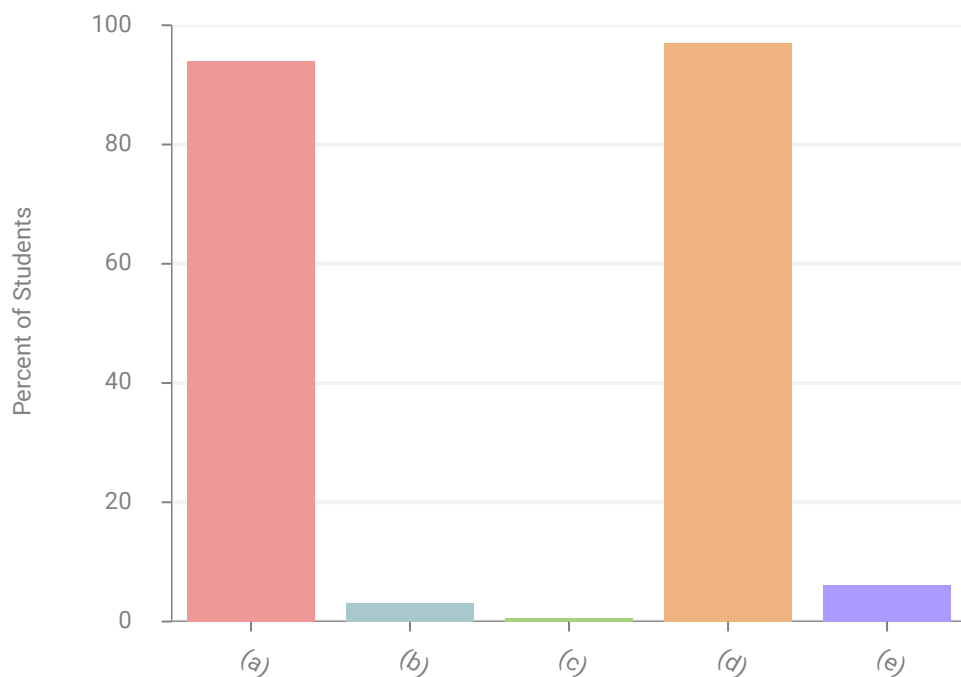


(d)



(e)





aninconle 2026-09-11 20:43:40

The answers are (a) and (d). Emilie is moving right in (a), so her velocity is $v_x = 6.6 \text{ m/s}$, but in (d) she's moving left, and her velocity is $v_x = -6.6 \text{ m/s}$.

aninconle 2026-09-11 20:43:41

Do you notice anything about Emilie's height at (a) and (d)?

Marshall_Huang 2026-09-11 20:44:55

They have the same height

AN29 2026-09-11 20:44:55

the height is the same

raphaelyuan 2026-09-11 20:44:55

the same

clairexc 2026-09-11 20:44:55

They are the same

Brightgiraffe 2026-09-11 20:44:55

they are about the same?

TatuShark 2026-09-11 20:44:55

they're the same?

aahelic13 2026-09-11 20:44:55

they are basically the same

Oceanix 2026-09-11 20:44:55

they look like they are the same

guacataj 2026-09-11 20:44:55

they are the same

kabir_sahil 2026-09-11 20:44:55

same

rysaishi 2026-09-11 20:44:55

It's the same

CodeWrecker 2026-09-11 20:44:55

they are the same or extremely similar

hw_zasf.xAx 2026-09-11 20:44:55

they are about the same

LCLARK5691 2026-09-11 20:44:55

It's the same!

Diamond_Zombie67 2026-09-11 20:44:55

they're the same

CAAI 2026-09-11 20:44:55

the same

flybirdeee_ 2026-09-11 20:44:55

same height

kanishka.chittoor 2026-09-11 20:44:55

same height

AllenLuo 2026-09-11 20:44:55

same height

Zip-a-Dee-Doo-Dah 2026-09-11 20:44:55

I have a feeling that were are about to learn something very cool.

Bunnicorn 2026-09-11 20:44:55

they're the same height

hhatchell 2026-09-11 20:44:55

They're the same?

WildOctopus 2026-09-11 20:45:04

they are equal

VeryDragon2013 2026-09-11 20:45:04

They're equal

Potatomiaomi 2026-09-11 20:45:04

the same

aninconle 2026-09-11 20:45:06

It's the same — one more clue that speed and height are related.

aninconle 2026-09-11 20:45:08

We can use t , a time coordinate, to measure time.

aninconle 2026-09-11 20:45:11

To find Emilie's average velocity over a short time, we could find her position at two times, t_1 and t_2 .

aninconle 2026-09-11 20:45:13

Emilie's coordinate at time t_1 can be x_1 . We'll use x_2 for her coordinate at time t_2 .

aninconle 2026-09-11 20:45:15

How would you compute how far Emilie moved between times t_1 and t_2 ?

Thorongil 2026-09-11 20:46:10

the displacement is the change in the x -coordinate, or $x_2 - x_1$

aninconle 2026-09-11 20:46:12

We'd take the difference in her coordinates, $x_2 - x_1$. This distance is called Emilie's *displacement*. A displacement is the change in position.

aninconle 2026-09-11 20:46:15

How do we find her average velocity over that time period?

Exponent11 2026-09-11 20:47:25

displacement divided by time

apollo2000 2026-09-11 20:47:25

divide the displacement over the time

M.Zhang555 2026-09-11 20:47:25

divide displacement by total time elapsed

VeryDragon2013 2026-09-11 20:47:25

$$\frac{x_2 - x_1}{t_2 - t_1}$$

Thorongil 2026-09-11 20:47:25

divide the displacement by the time, or $\frac{x_2 - x_1}{t_2 - t_1}$

Oceanix 2026-09-11 20:47:25

$(x_2 - x_1) / (t_2 - t_1)$

raphaelyuan 2026-09-11 20:47:25

$(x_2 - x_1) / (t_2 - t_1)$

Seifr 2026-09-11 20:47:25

$(x_2 - x_1) / (t_2 - t_1)$

aninconle 2026-09-11 20:47:33

We divide by how much time had passed, which is $t_2 - t_1$.

aninconle 2026-09-11 20:47:34

Emilie's average velocity is

$$v_{x;\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1}.$$

aninconle 2026-09-11 20:47:44

Here, there are two subscripts on $v_{x;\text{avg}}$. The x tells us it's a velocity along the x coordinate. The avg says it's an average. The semicolon keeps the subscripts separate.

aninconle 2026-09-11 20:47:46

It's important to keep in mind that this is a *model* of Emilie's motion. Can you think of anything about the motion of a real life skateboarder that is left out when we describe their position using just one coordinate?

guacataj 2026-09-11 20:50:00

i mean in real life there are 3 planes and air resistance and all that

aninconle 2026-09-11 20:50:03

A real skateboarder's entire body moves around as they ride. We don't include any such details in our model here. We simply think of the skateboarder as a single point.

aninconle 2026-09-11 20:50:05

In some cases, we'll need to build a more complicated model to include motion that is left out when we represent a real object as just a single coordinate. In other cases, the single coordinate is enough. There's no set of rules I can give you on when to use what model. This is a matter of judgment that you'll build as you examine more scenarios and try to build models yourself.

aninconle 2026-09-11 20:50:08

Back to Emilie's velocity and the formula

$$v_{x;\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1}.$$

aninconle 2026-09-11 20:50:10

If we switch from the x to the x' coordinate system, which things would change?

Emilie's coordinates

Emilie's position

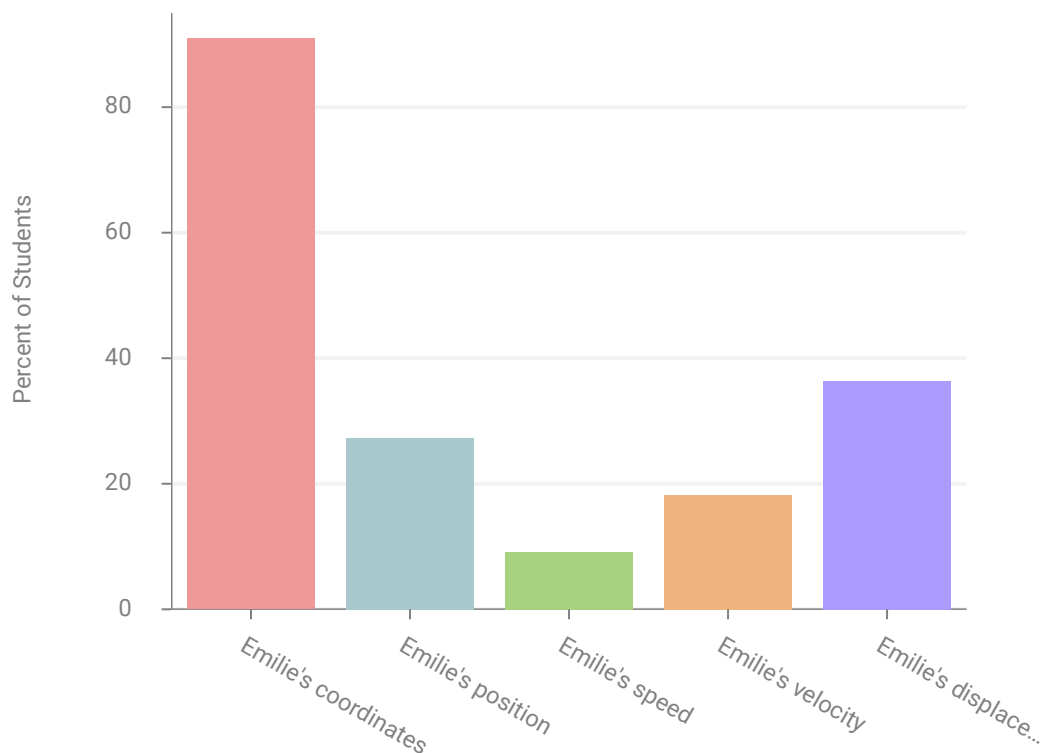
Emilie's speed

Emilie's velocity

Emilie's displacement

aninconle 2026-09-11 20:51:12

If we switch from the x to the x' coordinate system, which things would change?



aninconle 2026-09-11 20:51:17

Emilie's coordinates change, but her position, speed, velocity, and displacement do not.

aninconle 2026-09-11 20:51:20

Because displacement doesn't change when we change coordinate systems, it might be nice to give it some new bit of notation, meaning write it in some special way.

aninconle 2026-09-11 20:51:24

Although $x_2 - x_1$ is acceptable notation for displacement, we'll often use the notation Δx for it instead.

aninconle 2026-09-11 20:51:31

Δ is the Greek letter "Delta." Δx is pronounced "delta x." The Δ symbol means "change in."

aninconle 2026-09-11 20:51:34

How can we write an equation for Emilie's velocity using the Δ notation?

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

AllenLuo 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

AN29 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

pandafrog 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Bunicorn 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

apollo2000 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Oceanix 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

thekitsune 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Zip-a-Dee-Doo-Dah 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

aahelic13 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Potatomiaomi 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

hw_zasf.xAx 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Diamond_Zombie67 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

CodeWrecker 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Marshall_Huang 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

TatuShark 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

kanishka.chittoor 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

ndmath324458 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

syl314 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

kabir_sahil 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

WildOctopus 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

loripachiardi 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

CAAI 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Exponent11 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

M.Zhang555 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Ash_Ketchup13 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

agoel2010 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

guacataj 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

clairexc 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

flybirdeee_ 2026-09-11 20:53:16

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

raphaelyuan 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Thorongil 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Seifr 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

VeryDragon2013 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Nebula233 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

LCLARK5691 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Canada1334 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

Silverspoon 2026-09-11 20:53:32

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}$$

aninconle 2026-09-11 20:53:34

The time $t_2 - t_1$ can be written Δt , so

$$v_{x;\text{avg}} = \frac{\Delta x}{\Delta t}.$$

aninconle 2026-09-11 20:53:36

This still doesn't give us Emilie's *instantaneous* velocity, only her average velocity over some time. To work towards the idea of an instantaneous velocity, we'll need to bring in one more tool: graphing. This is what we'll do next week, when we look at changing velocities and the idea of acceleration.

aninconle 2026-09-11 20:54:13

Alright! We are about out of time. At the end of every class, I will give a summary. At the end of this class, I will talk more about the class.

aninconle 2026-09-11 20:54:16

Summary

aninconle 2026-09-11 20:54:17

We learned a lot of new terms today. Here's a review, you're welcome to go over them one by one on the transcript, and go over the relevant areas of today's class for any terms you want extra practice on.

aninconle 2026-09-11 20:54:20

model: a simplified way of thinking about something, potentially including diagrams, equations, graphs, etc.

sense-making: checking our results to see if they are reasonable

uncertainty: the size of a window into which we think our results should fall

dot diagram: a set of dots dropped at regular intervals marking something's location over time

position: actual place that something is

coordinate: quantity we assign to a position, symbol x

coordinate system: a set of coordinates, one for each possible position

speed: how fast something is going, symbol v

velocity: like a speed, but positive for motion to the right and negative for motion to the left, symbol v_x , or v_y , etc.

magnitude: physics word for absolute value. Speed is the magnitude of velocity.

Δ : Greek Delta, used for "change in"

displacement: how far something has moved, symbol Δx

instantaneous velocity: velocity at a specific moment

aninconle 2026-09-11 20:54:22

Speed is about the relationship between distance traveled and time. When can interpret a scenario in terms of a distance traveled in a given time, we can model the motion with a speed.

aninconle 2026-09-11 20:54:25

An object's *speed* is how fast it's going, and its *velocity* combines that with a direction of motion. We can use $v_{x;avg} = \frac{\Delta x}{\Delta t}$ to calculate a velocity.

aninconle 2026-09-11 20:54:34

We can use mathematics to build models of motion. Those models have limitations, but allow us to answer specific questions about the motion that we might not be able to answer without the models.

aninconle 2026-09-11 20:54:36

When we do modeling, we should also engage in sense-making to see if our model is reasonable. One way to do that is to check some particular values to see if they are approximately correct. Another is to think about extreme cases, where we can often work out what a model should do, even if we don't know what it does in every intermediate case.

aninconle 2026-09-11 20:54:46

Now, let's talk a little bit more about what you should do between classes. The homework is crucial for your learning; you'll reinforce what you learned today and investigate concepts more deeply.

aninconle 2026-09-11 20:54:49

First, if you haven't been to the course homepage yet, check it out right after class. You can get to it by clicking "My Classes" when you're logged in to the AoPS website.

aninconle 2026-09-11 20:54:51

Overview

This tab includes important introductory information and the lesson schedule. I'll post transcripts of all the class material here after each lesson.

aninconle 2026-09-11 20:54:53

Have you come up with a plan for when to do your homework?

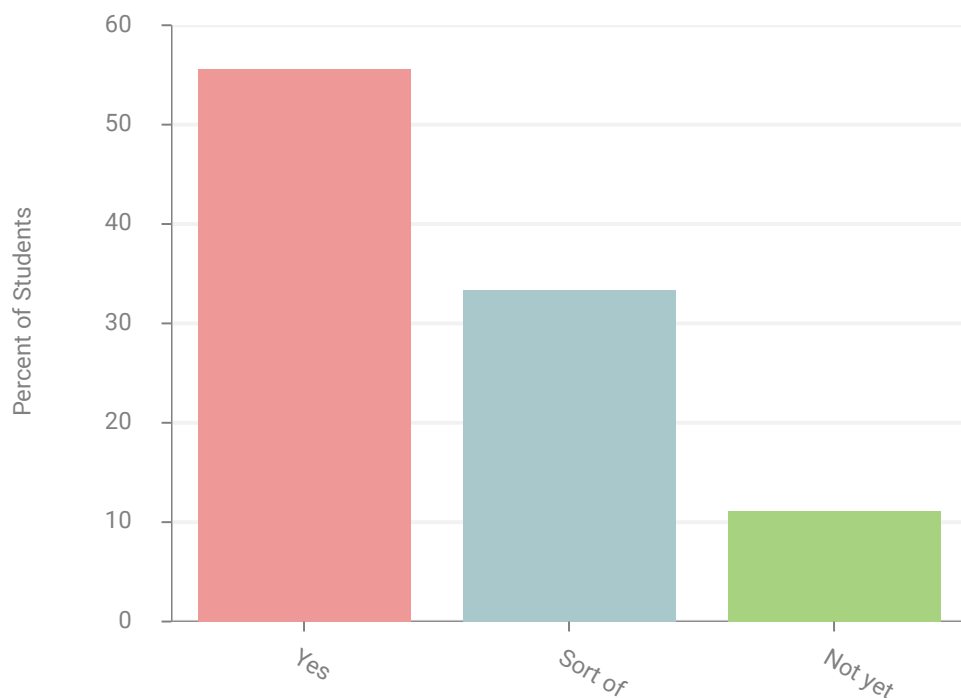
I know when and where I intend to study.

I have a vague idea.

I haven't thought about it yet, but making a plan is a good idea!

aninconle 2026-09-11 20:55:25

Have you come up with a plan for when to do your homework?



aninconle 2026-09-11 20:55:26

My Goals

Once you enter the days you want to study, this tab will suggest when to complete each task. Following this schedule will help you be successful in class without cramming. The My Goals tab will check some items off the list automatically.

aninconle 2026-09-11 20:55:28

Homework

This tab includes problems to solve between lessons. We provide full solutions to every problem. The short-answer challenge problems will be graded by the computer as soon as you submit your answer.

aninconle 2026-09-11 20:55:32

To answer a Writing Problem, write a full solution to the challenge problem using words and notation. It's best to write out your reasoning in a way that you think one of your classmates could understand! Whatever you have saved will be automatically collected the evening the homework is due and then graded by AoPS staff within a week. If you run out of time, you can click "Request Extension" on the Homework tab to ask for more time.

aninconle 2026-09-11 20:55:40

Prep Work

There's an interactive pre-reading before each class, found in the Prep Work tab. It's important to work through the reading before each class because we will begin using ideas from the pre-reading right away.

aninconle 2026-09-11 20:55:41

Unlike other AoPS classes, this prep work is not from a book. Instead, you'll have a dialogue with the computer, and you'll get to answer questions and play with simulations right there in the prereading.

aninconle 2026-09-11 20:55:43

Report

This tab will tell you how you're doing in this course. Report bars track your performance in class and on the homework.

aninconle 2026-09-11 20:55:45

How can you check if you've understood the material without solving every problem?

I can move on when every report bar is or .

red

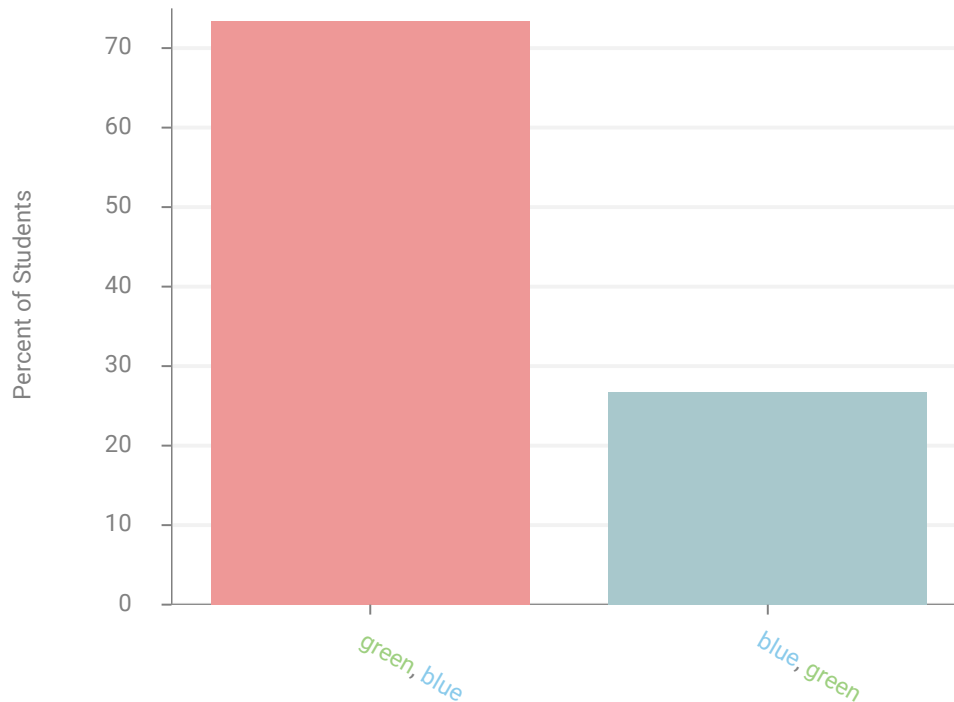
orange

green

blue

aninconle 2026-09-11 20:56:31

How can you check if you've understood the material without solving every problem?



aninconle 2026-09-11 20:56:33

Green bars mean that you've passed the task and can move on. Blue bars mean you've mastered the task.

aninconle 2026-09-11 20:56:34

Red bars and orange bars mean "Keep going!" Either solve more problems or review the problems you solved incorrectly to make sure you know how to solve them now.

aninconle 2026-09-11 20:56:36

The problems are intentionally difficult, and we don't expect you to be able to solve all of them on your first try. So we provide several resources to help you when you inevitably get stuck.

aninconle 2026-09-11 20:56:38

Message Board

The Message Board is a great place to ask questions about course content or socialize with your classmates. If you have questions about course logistics, please email success@aops.com instead.

aninconle 2026-09-11 20:56:39

Find the Message Board tab in your course homepage, or use the direct link: aops.com/class/4926/forum.

aninconle 2026-09-11 20:56:46

The Message Board is where you should go when you want help on the homework. The homework problems in this course are challenging, so you probably will want some help at some point!

aninconle 2026-09-11 20:56:47

We have integrated the message board into the transcripts and homework to make it easier to discuss course materials -- even specific lines in the transcript or problems in the homework. If you see a speech balloon icon, , clicking it will show you the threads already discussing that item.

aninconle 2026-09-11 20:56:49

Click a pencil and paper icon, , to create a new thread linked to that item. A linked thread will make it easier for everyone to see exactly what you want to discuss. If you've already solved a challenge problem and read the solution, click  to create a new thread invisible to students who have not yet solved the problem.

aninconle 2026-09-11 20:56:52

You have the option to ask questions that are anonymous to the other students (though you needn't ever be embarrassed to ask a question!).

aninconle 2026-09-11 20:56:58

AoPS staff are online throughout the day. Ask a question on the message board at any time, and someone will help you with your question within a business day.

aninconle 2026-09-11 20:57:01

To answer a Discussion Problem, respond to the prompt on the message board and discuss it with your classmates. Remember to be constructive, kind, and on-topic. If the problem is open-ended, you can just share your thoughts instead of coming up with a full solution. Moderators will be checking these threads to help keep the conversation on-track and provide feedback when necessary. Make sure to contribute to the discussion at least once.

aninconle 2026-09-11 20:57:04

Every lesson, you'll have either a Discussion Problem or a Writing Problem. We will alternate between discussion problems and writing problems. **There is a Writing Problem for the first lesson** and all odd-numbered lessons. Even-numbered lessons have Discussion Problems. These problems help you build vital communication skills, so make sure to start them as soon as possible.

aninconle 2026-09-11 20:57:12 

Credits

rental car story: [https://toronto.ctvnews.ca/woman-charged-8-000-after-rental-car-company-claims-she-drove-36-000-kilometres-in-three-days-1.6042079?fbclid=IwAR3ApAfv8br8uliQEZZ6rpwKI-oBxzs8Ts7B_cLkKZ4iUqwaUjpNXxusKJwAyesha McGowan: BBCNews, "Meet the African American woman cycling pioneer," https://www.youtube.com/watch?v=oHdXPwySS1w](https://toronto.ctvnews.ca/woman-charged-8-000-after-rental-car-company-claims-she-drove-36-000-kilometres-in-three-days-1.6042079?fbclid=IwAR3ApAfv8br8uliQEZZ6rpwKI-oBxzs8Ts7B_cLkKZ4iUqwaUjpNXxusKJwAyesha McGowan: BBCNews,)

wheelbase: By Rishiyur1 - This file has been extracted from another file, Public Domain,

<https://commons.wikimedia.org/w/index.php?curid=84331863>

pixels: By ed g2s • talk - Example image is a rendering of Image: Personal computer, exploded 5.svg., CC BY-SA 3.0,

<https://commons.wikimedia.org/w/index.php?curid=807503>

Fizeau schematic: By Brews ohare - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=6229074>

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aninconle 2026-09-11 20:57:23

Alright, any questions on how the class works?

M.Zhang555 2026-09-11 20:57:37

do we have tests

aninconle 2026-09-11 20:57:43

No tests, just homework.

Oceanix 2026-09-11 20:57:48

Where do i put the answers to the questions

aninconle 2026-09-11 20:58:04

You should be able to submit them in the homework. If that is what you mean.

kabir_sahil 2026-09-11 20:58:12

how do discussion problems workz/

kabir_sahil 2026-09-11 20:58:12

do we do them in class?

aninconle 2026-09-11 20:58:21

We won't be doing them in class.

aninconle 2026-09-11 20:58:35

You job there is just to talk about the problem.

aninconle 2026-09-11 20:59:35

Alright, have fun!



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